PR-PL: A Novel Prototypical Representation Based Pairwise Learning Framework for Emotion Recognition Using EEG Signals

Rushuang Zhou, Zhiguo Zhang, *Member, IEEE*, Hong Fu, Li Zhang, Linling Li, Gan Huang, Fali Li, Xin Yang, Yining Dong, Yuan-Ting Zhang, *Fellow, IEEE*, and Zhen Liang, *Member, IEEE*

***Abstract*—Affective brain-computer interface based on electroencephalography (EEG) is an important branch in the field of affective computing. However, the individual differences in EEG emotional data and the noisy labeling problem in the subjective feedback seriously limit the effectiveness and generalizability of existing models. To tackle these two critical issues, we propose a novel transfer learning framework with Prototypical Representation based Pairwise Learning (*PR-PL*). The discriminative and generalized EEG features are learned for emotion revealing across individuals and the emotion recognition task is formulated as pairwise learning for improving the model tolerance to the noisy labels. More specifically, a prototypical learning is developed to encode the inherent emotion-related semantic structure of EEG data and align the individuals' EEG features to a shared common feature space under consideration of the feature separability of both source and target domains. Based on the aligned feature representations, pairwise learning with an adaptive pseudo labeling method is introduced to encode the proximity relationships among samples and alleviate the label noises effect on modeling. Extensive results on two benchmark databases (SEED and SEED-IV) under four different cross-validation evaluation protocols validate the model reliability and stability across subjects and sessions. Compared to the literature, the average enhancement of emotion recognition across four different evaluation protocols is 2.04% (SEED) and 2.58% (SEED-IV).**

***Index Terms*—EEG, emotion recognition, affective brain-computer interface, transfer learning, affective computing.**

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Rushuang Zhou is with the School of Biomedical Engineering, Medical School, Guangdong Provincial Key Laboratory of Biomedical Measurements and Ultrasound Imaging, Shenzhen University, Shenzhen 518060, China, and also with the Department of Biomedical Engineering, Hong Kong Center for Cerebro-Cardiovascular Health Engineering (COCHE), Hong Kong Science and Technology Park, City University of Hong Kong, Kowloon Tong, Hong Kong (e-mail: rrushuang2-c@my.cityu.edu.hk).

Zhiguo Zhang is with the Institute of Computing and Intelligence, Harbin Institute of Technology, Shenzhen 150001, China, and also with Peng Cheng Laboratory, Shenzhen 518066, China (e-mail: zhiguozhang@hit.edu.cn).

Hong Fu is with the Department of Mathematics and Information Technology, The Education University of Hong Kong, Ting Kok, Hong Kong (e-mail: hfu@eduhk.hk).

Li Zhang, Linling Li, Gan Huang, Xin Yang, and Zhen Liang are with the School of Biomedical Engineering, Medical School, Guangdong Provincial Key Laboratory of Biomedical Measurements and Ultrasound Imaging, Shenzhen University, Shenzhen 518060, China (e-mail: lzhang@szu.edu.cn; lilinling@szu.edu.cn; huanggan@szu.edu.cn; yangxinknow@gmail.com; zhenliang.szu@gmail.com).

Fali Li is with the The Clinical Hospital of Chengdu Brain Science Institute, MOE Key Lab for Neuroinformation, School of Life Science and Technology, Center for Information in Medicine, University of Electronic Science and Technology of China, Chengdu 610056, China (e-mail: fali.li@uestc.edu.cn).

Yining Dong is with the School of Data Science, Hong Kong Center for Cerebro-Cardiovascular Health Engineering (COCHE), Hong Kong Science and Technology Park, City University of Hong Kong, Kowloon Tong, Hong Kong (e-mail: yinidong@cityu.edu.hk).

Yuan-Ting Zhang is with the Department of Biomedical Engineering, Hong Kong Center for Cerebro-Cardiovascular Health Engineering (COCHE), Hong Kong Science and Technology Park, City University of Hong Kong, Kowloon Tong, Hong Kong (e-mail: yt.zhang@cityu.edu.hk).

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I. Introduction

AFFECTIVE computing is a fast growing interdisciplinary research field and is attracting researchers' attention from different areas including computer science, neuroscience, psychology, and signal processing [1]. Recently, electroencephalography (EEG) based emotion recognition has become an increasingly important topic for affective computing and human sentiment analysis [2], [3]. A proper design of EEG-based emotion recognition models is helpful for facilitating data processing, benefiting discriminant feature characterization, and lightening model performance. Currently, there exist two main critical issues in EEG-based emotion recognition models. One is *individual differences*: how to build a generalized affective computing model which could adapt to the remarkable individual differences and detect discriminative features in the simultaneously collected EEG signals; and another is *noisy labeling*: how to build a reliable and stable affective computing model which is more tolerance to the contaminated label information.

PR-PL：一种基于原型表示的成对学习框架，用于脑电图（EEG）信号的情绪识别

周如霜，张志国，IEEE 会员，付红，张丽，李林林，黄干，李发利，杨欣，董一宁，IEEE 会士张元庭，以及梁振，IEEE 会员

摘要—基于脑电图（EEG）的情感脑机接口是情感计算领域的一个重要分支。然而，脑电图（EEG）中的个体差异情感数据以及主观反馈中的噪声标签问题严重限制了现有模型的有效性和泛化能力。为了解决这两个关键问题，我们提出了一种基于原型表示的成对学习（PR-PL）的新型迁移学习框架。该框架学习具有区分性和泛化能力的脑电图（EEG）特征，以揭示个体间的情感差异，并将情感识别任务表述为成对学习，以提高模型对噪声标签的容忍度。具体而言，我们开发了一种原型学习方法，用于编码 EEG 数据的内在情感相关语义结构，并在考虑源域和目标域特征可分性的情况下，将个体的 EEG 特征对齐到一个共享的特征空间中。基于对齐的特征表示，我们引入了一种具有自适应伪标签方法的成对学习，用于编码样本之间的邻近关系，并减轻标签噪声对建模的影响。 在两个基准数据库（SEED 和 SEED-IV）上，通过四种不同的交叉验证评估协议获得的大量结果验证了该模型在受试者和会话中的可靠性和稳定性。与文献相比，在四种不同的评估协议下，情绪识别的平均提升分别为 2.04%（SEED）和 2.58%（SEED-IV）。

关键词—脑电图（EEG）、情绪识别、情感脑机接口、迁移学习、情感计算。

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Rushuang Zhou 是深圳大学医学院生物医学工程学院、广东省生物医学测量与超声成像重点实验室的成员，同时也是香港城市大学香港脑心健康工程中心（COCHE）生物医学工程学院的成员，地址为中国深圳 518060 和香港九龙塘香港科学园（邮箱：rrushuang2-c@my.cityu.edu.hk）。

Zhiguo Zhang 是哈尔滨工业大学计算机与智能学院、深圳 150001 中国的成员，同时也是鹏城实验室、深圳 518066 中国的成员（邮箱：zhiguozhang@hit.edu.cn）。

Hong Fu 是香港教育大学数学与信息技术学院的成员，地址为香港天水围（邮箱：hfu@eduhk.hk）。

李张，李林林，黄干，杨欣，梁振与深圳大学生物医学工程学院、医学院、广东省生物医学测量与超声成像重点实验室合作，地址为中国深圳市 518060（邮箱：lzhang@szu.edu.cn；lilinling@szu.edu.cn；huangan@szu.edu.cn；yangxinknow@gmail.com；zhenliang.szu@gmail.com）。

李发利与电子科技大学成都脑科学研究所临床医院、教育部神经信息重点实验室、生命科学与技术学院、医学信息中心合作，地址为中国成都市 610056（邮箱：fali.li@uestc.edu.cn）。

董一宁与香港城市大学数据科学学院、香港脑心健康工程中心（COCHE）、香港科学园合作，地址为香港九龙塘（邮箱：yinidong@cityu.edu.hk）。

张元庭与香港城市大学生物医学工程学院、香港脑心健康工程中心（COCHE）、香港科学园合作，地址为香港九龙塘（邮箱：yt.zhang@cityu.edu.hk）。

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I. I

情感计算是一个快速发展的跨学科研究领域，吸引了来自计算机科学、神经科学、心理学和信号处理等不同领域研究人员的关注[1]。近年来，基于脑电图（EEG）的情绪识别已成为情感计算和人类情感分析日益重要的课题[2],[3]。合理设计基于 EEG 的情绪识别模型有助于促进数据处理、有利于判别特征表征，并提升模型性能。目前，基于 EEG 的情绪识别模型存在两个主要关键问题。一是个体差异：如何构建一个通用的情感计算模型，使其能够适应显著的个体差异，并在同时收集的 EEG 信号中检测判别特征；二是噪声标签：如何构建一个可靠且稳定的情感计算模型，使其更能容忍被污染的标签信息。

Recently, more and more researchers have focused on applying transfer learning methods to alleviate the individual differences in EEG signals [4], [5], [6], [7], [8], [9] and improve feature invariant representation [10], [11], [12]. Under consideration of the feature distribution of both the source domain (labeled data with known distribution) and the target domain (unlabeled data with unknown distribution), the transfer learning methods try to minimize the distribution difference between the source domain and the target domain by approximately satisfying the assumption of independent and identical distribution and can consequently realize a higher recognition performance on the target domain. Through a domain-shifting strategy, the invariant feature representations across different domains are learned and the relationships between the learned features and emotional labels are explored. Currently, most existing EEG-based emotion recognition models are based on the transfer learning framework, in which both the source domain (data and label) and the target domain (data only) are seen during model training. For example, Li et al. [5] proposed a multisource transfer learning method with two different stages. In the first stage, appropriate samples were selected from the existing source domain. In the second stage, a style transfer mapping was implemented to alleviate the differences between the selected source samples and the unknown target samples. The results showed the proposed transfer method outperformed the non-transfer method with an improvement of 12.72% on the public SEED database [13] with a three-class classification problem (negative, neutral, and positive). Inspired by neuroscience findings that different emotions would lead to different brain reactions, Li et al. [6] proposed a novel R2G-STNN network to integrate the EEG spatial-temporal dynamics at the local and global brain areas and realize an efficient emotion recognition performance together with a domain shift learning. More details about current EEG-based emotion recognition models with the transfer learning algorithms are presented in Section II.

For video-evoking EEG-emotion experiments, subjects may not always be able to accurately react to the intended emotions, and at the same time may not be able to accurately describe and feedback on their emotional changes. This would bring label noise to the emotional information annotation of EEG samples and further lead to a negative impact on the model performance [14]. To tackle this issue, Zhong et al. [8] developed an emotion-aware distribution learning method (RGNN), in which they blurred the label information by changing the one-hot label representation $(1, 0, 0)$ to $(1 - \frac{2\epsilon}{3}, \frac{2\epsilon}{3}, 0)$ and trained the model to be less sensitive to the label noise. However, the model performance would greatly rely on the selection of ϵ value, and an optimal ϵ value selection could be different for different databases and different individuals. On the other hand, current EEG-based emotion recognition models are mainly realized by *pointwise learning*, which trains a model to be close to the given labels ($y = f(x)$). In other words, if the given labels are fully contaminated by noises, the model learning process could be easily affected. Contrarily, *pairwise learning* is to train a model to learn the relative associations between pairs of instances ($f(x_i, x_j)$) and encode the proximity among samples with less reliance on labeling. Thus, pairwise learning has achieved tremendous success in tackling the noisy labeling issue in a number of real-world applications [15], [16], [17], [18].

In this paper, we formulate the EEG-based emotion recognition tasks as a pairwise learning problem and propose a novel transfer learning framework with prototypical representation based pairwise learning (which is termed as *PR-PL* below). Here, we model the relative relationship between pairs of EEG samples in terms of prototypical representations, which is advantageous to pointwise learning especially when the high-precision labels are difficult to obtain [19]. The major novelties of the proposed PR-PL are summarized as follows. (1) We propose a novel transfer learning framework to tackle the individual differences and noisy labeling issues which commonly limit the model capacity in the current EEG-based emotion recognition models. Different from the existing transfer learning frameworks that only focus on feature separability on the source domain, we introduce an unsupervised adaptive pseudo labeling method to generate pseudo labels for the target domain and consider the feature separability in both the source domain and the target domain through an end-to-end domain adversarial training, where the model effectiveness and generalizability could be further enhanced. (2) We develop a prototypical learning-based adversarial discriminative domain adaptation method to explore latent variables of EEG signals corresponding to emotion categories, encode the semantic structure of EEG data, and learn subject-generalized prototypical representations for emotion representation across individuals. The characterized prototypical representations show a high feature concentration within one single emotion category and a high feature separability across different emotion categories. (3) We first formulate emotion recognition as a pairwise learning problem to replace the traditional classifier and greatly alleviate the label dependence on high-precision emotion labels. The pairwise learning provides us with an alternative way to interpret the EEG signals belonging to the same emotion category under a noisy labeling environment.

II. Related Work

The existing EEG-based emotion recognition models with the transfer learning algorithms can be generally divided into two main categories.

(a) Non-Deep Transfer Learning Models: Pan et al. [20] proposed a transfer component analysis (TCA) algorithm to reduce the marginal distribution difference between the source and target domains, in which the transfer information was learned in a reproducing kernel Hilbert space through maximizing mean discrepancy. Zheng and Lu [21] introduced two types of subject-to-subject transfer methods to deal with the challenge of individual differences in EEG signal processing. One was to explore a shared common feature space underlying source and target domains using TCA and kernel principal analysis (KPCA), and another was to construct multiple personalized classifiers on the source domain and map the classifier parameters to the target domain using transductive parameter transfer (TPT). These non-deep transfer learning strategies have shown the possibility to bridge the discrepancy across two domains

近年来，越来越多的研究人员开始关注将迁移学习方法应用于缓解脑电图（EEG）信号中的个体差异[4], [5], [6], [7], [8], [9]并提升特征不变表示[10], [11], [12]。在考虑源域（具有已知分布的标记数据）和目标域（具有未知分布的无标记数据）的特征分布的情况下，迁移学习方法通过近似满足独立同分布的假设来最小化源域和目标域之间的分布差异，从而能够在目标域上实现更高的识别性能。通过领域迁移策略，学习到跨不同域的不变特征表示，并探索学习到的特征与情绪标签之间的关系。目前，大多数现有的基于 EEG 的情绪识别模型都基于迁移学习框架，在该框架中，模型训练时会同时看到源域（数据和标签）和目标域（仅数据）。例如，Li 等人。 [5] 提出了一种具有两个不同阶段的多源迁移学习方法。在第一阶段，从现有源域中选择合适的样本。在第二阶段，实施了一种风格迁移映射，以减轻所选源样本与未知目标样本之间的差异。结果表明，所提出的迁移方法在具有三类分类问题（消极、中性和积极）的公共 SEED 数据库[13]上优于非迁移方法，提高了 12.72%。受神经科学发现不同情绪会导致不同大脑反应的启发，Li 等人[6]提出了一种新的 R2G-STNN 网络，以整合局部和全局脑区的脑电时空动态，并实现高效的结合域迁移学习的情绪识别性能。关于当前基于脑电的情绪识别模型与迁移学习算法的更多细节，将在第二节中介绍。

对于视频诱发脑电情绪实验，受试者可能无法总是准确地对所期望的情绪做出反应，同时可能也无法准确描述和反馈其情绪变化。这会给脑电样本的情绪信息标注带来标签噪声，并进一步对模型性能产生负面影响[14]。为解决这一问题，Zhong 等人[8]开发了一种情绪感知分布学习方法（RGNN），其中他们通过将 one-hot 标签表示（1，0，0）改为（1 − ε，ε，0）来模糊标签信息，并训练模型使其对标签噪声不那么敏感。然而，模型性能会很大程度上依赖于ε值的选择，而最佳ε值的选择可能因不同的数据库和不同个体而异。另一方面，当前的基于脑电的情绪识别模型主要通过点式学习实现，即训练一个模型使其接近给定的标签（y = f(x)）。换句话说，如果给定的标签完全被噪声污染，模型的学习过程可能会很容易受到影响。 相反，成对学习旨在训练一个模型来学习实例对之间的相对关联（f(x, x)），并以较少依赖标签的方式编码样本间的邻近性。因此成对学习在解决多个实际应用中的噪声标签问题方面取得了巨大成功[15]，[16]，[17]，[18]。

在本文中，我们将基于脑电图（EEG）的情绪识别任务建模为成对学习问题，并提出了一种基于原型表示的成对学习迁移学习框架（以下简称 PR-PL）。在此框架中，我们通过原型表示来建模 EEG 样本对之间的相对关系，这对于点式学习尤其有利，尤其是在难以获得高精度标签的情况下[19]。所提出的 PR-PL 的主要创新点总结如下。(1)我们提出了一种新的迁移学习框架，以解决个体差异和噪声标签问题，这些问题通常限制当前基于 EEG 的情绪识别模型的能力。与仅关注源域特征可分性的现有迁移学习框架不同，我们引入了一种无监督自适应伪标签方法来为目标域生成伪标签，并通过端到端的域对抗训练来考虑源域和目标域中的特征可分性，从而进一步提升模型的有效性和泛化能力。 (2) 我们开发了一种基于原型学习的对抗判别域适应方法，用于探索与情绪类别相对应的脑电图信号的潜在变量，编码脑电图数据的语义结构，并学习跨个体的通用原型表示以进行情绪表征。这些特征化的原型表示在单个情绪类别内表现出高度的特征集中性，并在不同情绪类别之间表现出高度的特征可分离性。(3) 我们首先将情绪识别表述为成对学习问题，以替代传统分类器，并极大地减轻对高精度情绪标签的依赖。成对学习为我们提供了一种在噪声标签环境下解释属于同一情绪类别的脑电图信号的方法。

II. RW 基于脑电图的现有情绪识别模型与迁移学习算法通常可分为两大主要类别。

(a) 非深度迁移学习模型：Pan 等人[20]提出了一种迁移成分分析（TCA）算法，以减少源域和目标域之间的边缘分布差异，其中通过最大化平均差异在再生核希尔伯特空间中学习迁移信息。郑和陆[21]引入了两种主体到主体的迁移方法来应对脑电图信号处理中个体差异的挑战。一种是利用 TCA 和核主成分分析（KPCA）探索源域和目标域下潜在的共同特征空间，另一种是在源域构建多个个性化分类器，并使用传递参数迁移（TPT）将分类器参数映射到目标域。这些非深度迁移学习策略显示出弥合两个域之间差异的可能性

with improved performance on the target domain. However, due to the small capacity and low complexity of modeling, the model accuracy and stability are still constrained, failing to meet the needs of affective brain-computer interfaces (aBCIs) in real-world applications.

(b) Deep Transfer Learning Models: Most of the existing EEG-based emotion recognition models are based on deep transfer learning methods built with domain-adversarial neural network (DANN) proposed in [22]. The main idea of DANN is to find a shared feature representation for source and target domains with indistinguishable distribution differences and also maintain the predictive ability of the estimated features on the source samples for a specific classification task. Jin et al. [23] was the first to introduce DANN in aBCIs. Benefiting from the powerful feature representation ability of deep networks and the high efficiency of adversarial learning in distribution adaptation, the results have demonstrated that DANN based aBCI system is superior to the other existing methods. The current aBCI systems could be considered as a collection of DANN-based models, which typically start from three directions to enhance the DANN performance in resolving the EEG-based emotion recognition tasks.

- *Incorporating the prior knowledge of neuroscience and brain anatomy with DANN:* Inspired by the neuroscience findings of the asymmetry property between the left and right hemispheres in emotional responses, Li et al. [24] proposed a bi-hemisphere domain adversarial neural network (BiDANN), in which a global and two local domain discriminators were designed to learn discriminant features from each cerebral hemisphere related to emotion perception and to improve the feature stability to the variation of different domains. The experiments on the SEED database demonstrated that BiDANN achieved higher emotion recognition performance than the original DANN. Considering the emotional responses from different brain regions would be varied, Li et al. [6] introduced a region-global spatial-temporal neural network (R2G-STNN) to integrate the spatial-temporal information from local and global brain regions under importance guidance and characterize hierarchical feature representations for modeling. Similarly, under an assumption that not all EEG channels are equally important in emotion recognition tasks, Du et al. [25] integrated attention mechanism, long-short-term memory (LSTM), and DANN to propose an attention-based LSTM with domain discriminator (ATDD-LSTM), in which the nonlinear relations among different EEG channels were characterized in a data-driven approach and the informative emotion-related channels were optimally selected.
- *Incorporating the probability distribution with DANN:* To deal with the training instability of DANN, Luo et al. [26] introduced Wasserstein generative adversarial network with gradient penalty (WGAN-GP) to narrow down the distance between the marginal probability distributions of different subjects. The results showed the model stability was improved and a better EEG-based cross-subject emotion recognition was achieved. However, WGAN-GP only considered the marginal distribution differences but ignored the joint distribution differences of different domains. To address this problem, Li et al. [27] introduced a joint domain adaptation network (JLNN), where the joint distribution adaptation (JDA) method was incorporated with a unified framework of task-invariant features (MDA) and task-specific features (CDA).
- *Incorporating the metric learning with DANN:* Wang et al. [28] incorporated an efficient prototype-based data representation to learn a more discriminative embedded feature space for EEG-based emotion recognition, in which the euclidean metric was adopted to assess the associations between the data samples and all the selected prototypes. In order to jointly estimate the feature separability and distribution alignment, Wang et al. [29] presented a prototype-based symmetric positive definite (SPD) matrix network to integrate DANN with the metric learning based on the Riemannian distance. Shen et al. [30] proposed a coupled projection transfer metric learning (CPTML) model to take the domain alignment and graph-based metric learning into account at the same time.

Although the above models have achieved higher accuracies compared to the original model with DANN in EEG-based emotion recognition tasks, there still exist three major technical challenges. First, the learned feature representation is susceptible to noise interference from both source and target domains and would further affect the model generalizability [31], [32]. Second, the current DANN-based models primarily concern the emotion classification loss on the source domain, which might easily lead to the over-fitting of source domain data and a decline in classification ability on target domain data. Third, the existing algorithms largely rely on a large amount of accurately labeled data. However, in practical EEG applications, it is difficult to collect high-precision labels for each single EEG trial.

III. Methodology

Suppose the EEG trials on the source domain $\mathbb{S}$ and target domain $\mathbb{T}$ are given as $D_s = \{X_s, Y_s\}$ and $D_t = \{X_t, Y_t\}$, where $\{X_s, Y_s\} = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$ and $\{X_t, Y_t\} = \{(x_i^t, y_i^t)\}_{i=1}^{N_t}$. Here, X_i^s and X_i^t are EEG samples, and Y_i^s and Y_i^t are the corresponding emotion labels. To make the narrative clearer, the frequently used notations are summarized in Table I. As shown in Fig. 1, the proposed PR-PL includes two main parts (prototypical representation and pairwise learning), with three loss functions (domain adversarial loss, supervised pairwise loss (source domain), and unsupervised pairwise loss (target domain)). In the prototypical representation, three types of features are defined. The characterized features $f(X_s)$ or $f(X_t)$ from the EEG samples are termed as *sample features*, which are forced to be as indistinguishable as possible from the source and target domains. Under an assumption that any emotion could be represented by a prototype via prototypical learning, the *prototype features* of each emotion category are learned based on the sample features $f(X_s)$ and Y_s from the source domain. The interaction relationships between the sample features and prototype features are measured and the *interaction features* are characterized which

在目标领域上性能得到提升。然而，由于建模能力有限且复杂性低，模型的准确性和稳定性仍然受限，无法满足实际应用中情感脑机接口（aBCIs）的需求。

(b) 深度迁移学习模型：现有的大多数基于脑电图（EEG）的情绪识别模型基于深度迁移学习方法构建，该方法采用了在[22]中提出的领域对抗神经网络（DANN）。DANN 的主要思想是找到具有不可区分分布差异的源域和目标域的共享特征表示，并保持对特定分类任务中源样本的估计特征的预测能力。金等人[23]首次将 DANN 引入到脑机接口（BCI）中。受益于深度网络强大的特征表示能力和对抗学习在分布适应中的高效率，结果表明基于 DANN 的 BCI 系统优于其他现有方法。当前的 BCI 系统可以被视为由基于 DANN 的模型集合组成，这些模型通常从三个方向出发来提升 DANN 在解决基于 EEG 的情绪识别任务中的性能。

脑结构与 DANN：受神经科学研究中左右半球在情绪反应中不对称性发现的启发，Li 等人[24]提出了一种双半球域对抗神经网络（BiDANN），其中设计了一个全局域判别器和两个局部域判别器，用于从与情绪感知相关的每个大脑半球中学习判别特征，并提高特征对不同域变化的稳定性。在 SEED 数据库上的实验表明，BiDANN 比原始 DANN 实现了更高的情绪识别性能。考虑到不同脑区的情绪反应会有所不同，Li 等人[6]引入了一种区域全局时空神经网络（R2G-STNN），在重要性引导下整合局部和全局脑区的时空信息，并表征层次特征表示以进行建模。类似地，在假设并非所有 EEG 通道在情绪识别任务中同等重要的前提下，Du 等人 [25] 集成了注意力机制、长短期记忆网络（LSTM）和判别式对抗神经网络（DANN），提出了基于注意力的 LSTM 与领域判别器（ATDD-LSTM），其中通过数据驱动方法刻画了不同 EEG 通道之间的非线性关系，并最优地选择了信息丰富的情绪相关通道。 融入 DANN 的概率分布：为解决 DANN 的训练不稳定性问题，Luo 等人

[26] 引入了 Wasserstein 生成对抗网络与梯度惩罚（WGAN-GP），以缩小不同受试者边际概率分布之间的距离。结果表明，模型稳定性得到提升，并实现了更好的基于 EEG 的跨受试者情绪识别。然而

WGAN-GP 仅考虑了边缘分布差异，而忽略了不同域的联合分布差异。为解决这一问题，Li 等人 [27] 引入了一个联合域适应网络（JLNN），将联合分布适应（JDA）方法与任务不变特征（MDA）和任务特定特征（CDA）的统一框架相结合。将度量学习与 DANN 结合：Wang 等人 [28] 引入了一种基于原型的有效数据表示方法，用于学习更具有判别性的嵌入特征空间，用于基于 EEG 的情绪识别，其中采用欧几里得度量来评估数据样本与所有选定原型之间的关联。为了联合估计特征可分性和分布对齐，Wang 等人 [29] 提出了一种基于原型的对称正定（SPD）矩阵网络，将 DANN 与基于黎曼距离的度量学习相结合。Shen 等人 [30] 提出了一种耦合投影转移度量学习（CPTML）模型，同时考虑了域对齐和基于图的度量学习。

尽管上述模型在基于脑电图（EEG）的情绪识别任务中，与原始的 DANN 模型相比实现了更高的准确率，但仍然存在三个主要的技术挑战。首先，学习到的特征表示容易受到源域和目标域噪声的干扰，这将进一步影响模型的泛化能力[31], [32]。其次，当前的基于 DANN 的模型主要关注源域的情绪分类损失，这很容易导致源域数据的过拟合，并降低目标域数据的分类能力。第三，现有的算法很大程度上依赖于大量精确标记的数据。然而，在实际的 EEG 应用中，很难为每个单一的 EEG 试验收集高精度的标签。

III. M 假设源域 S 和目标域 T 上的脑电图试验分别给定为 D= {X, Y}和 D= {X, Y}，其中{X, Y} = {(x, y)}i 和{X, Y} = {(x, y)}。这里，X 和 X 是脑电图样本，Y 和 Y 是对应的情绪标签。为了使叙述更清晰，常用的符号总结在表 I 中。如图 1 所示，所提出的 PR-PL 包括两个主要部分（原型表示和成对学习），以及三个损失函数（域对抗损失、监督成对损失（源域）和无监督成对损失（目标域））。在原型表示中，定义了三种类型的特征。从脑电图样本中提取的特征 f (X)或 f (X)被称为样本特征，这些特征被强制要求尽可能与源域和目标域区分不开。在假设任何情绪都可以通过原型学习表示为原型的前提下，基于源域的样本特征 f (X)和 Y，学习每个情绪类别的原型特征。 样本特征与原型特征之间的交互关系被测量，交互特征被表征

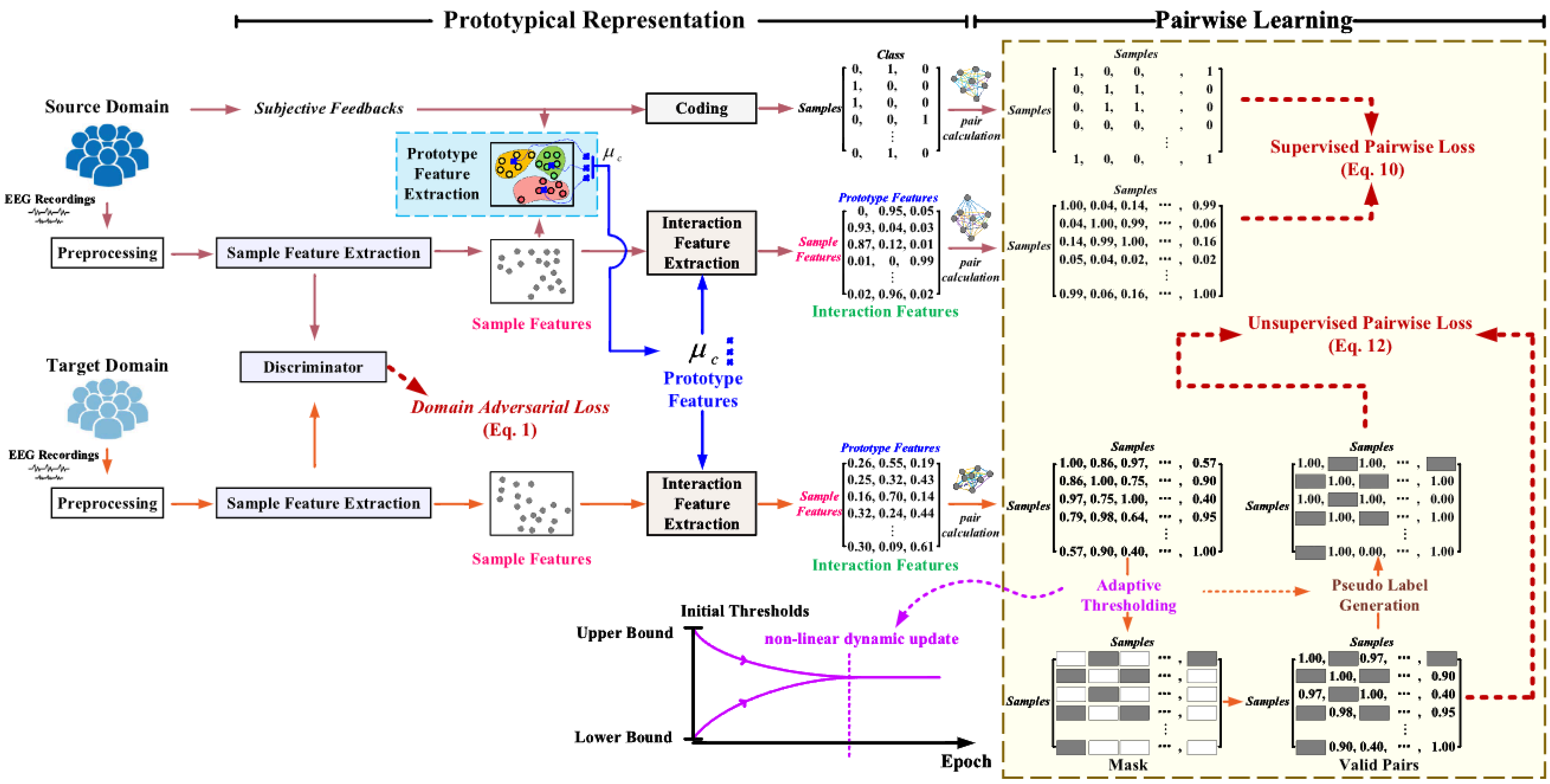

Fig. 1. The proposed PR-PL framework. In this proposed PR-PL, the domain adversarial loss, supervised pairwise loss, and unsupervised pairwise loss are defined in (1), (10), and (12), respectively. The final objective function is a combination of the above three loss functions, given in (17).

TABLE I
FREQUENTLY USED NOTATIONS AND DESCRIPTIONS

Notation	Description
$\mathbb{S}\backslash\mathbb{T}$	source\target domain
$x^s\backslash x^t$	source\target feature
$y^s\backslash y^t$	source\target label
$N^s\backslash N^t$	number of source\target samples
$D^s\backslash D^t$	the source\target dataset
$f(\cdot)$	sample feature extractor
$d(\cdot)$	domain discriminator
$h(\cdot)$	bilinear operation
μ_c	prototype features
l_i	interaction features
θ_f	the parameter of the feature extractor $f(\cdot)$
θ_d	the parameter of the discriminator $d(\cdot)$
S	bilinear transformation matrix in $h(\cdot)$

will be used in the following pairwise learning. In pairwise learning, the pair relationships on both source and target domains are explored. As the information about Y_t on the target domain is unseen during model training, an unsupervised pairwise learning strategy is proposed for pseudo labeling and mask generation with a non-linear dynamic thresholding method.

A. Sample Feature Extraction

To make both source and target domain data satisfy the assumption of independent and identical distribution and obey the same distribution, we characterize the sample features based on a domain adversarial training introduced in DANN. Here, the distribution difference between the source domain and the target domain is alleviated and the sample features with domain invariant properties are characterized. Through this process, the individual differences in EEG signals could be alleviated and the generalization ability of the models could be improved [6], [24], [25], [26]. Specifically, the domain difference between the sample feature $f(X_s)$ on the source domain and the sample feature $f(X_t)$ on the target domain are minimized by adopting domain adversarial training. Here, $f(\cdot)$ with the parameter θ_f is the designed feature extractor for extracting the sample features from EEG signals. A domain discriminator $d(\cdot)$ with the parameter θ_d is introduced to distinguish whether the characterized sample features ($f(X_s)$ or $f(X_t)$) come from the source domain ($\mathbb{S}$) or the target domain ($\mathbb{T}$), whose loss function is a standard two-category cross-entropy loss function, given as

$$\mathcal{L}_{disc}(\theta_f,\theta_d) = -\sum_{i=1}^{N_s}\log d\left(f\left(x_i^s\right)\right) - \sum_{i=1}^{N_t}\log\left(1-d\left(f\left(x_i^t\right)\right)\right), \quad (1)$$

which is termed as *domain adversarial loss* in Fig. 1.

In the training process, we adopt the end-to-end training method [22], and implement the domain adversarial training by introducing a gradient reversal layer. The feature extractor $f(\cdot)$ maximizes the classification ability to enhance emotion recognition performance and at the same time, the discriminator $d(\cdot)$ minimizes the domain discrimination to reduce the distribution difference between the source and target domains. The objective function in the domain adversarial training process could be represented as

$$\min_{\theta_f}\max_{\theta_d}\mathcal{L}_{\text{classifier}}(\theta_f) - \lambda\mathcal{L}_{\text{disc}}(\theta_f,\theta_d). \quad (2)$$

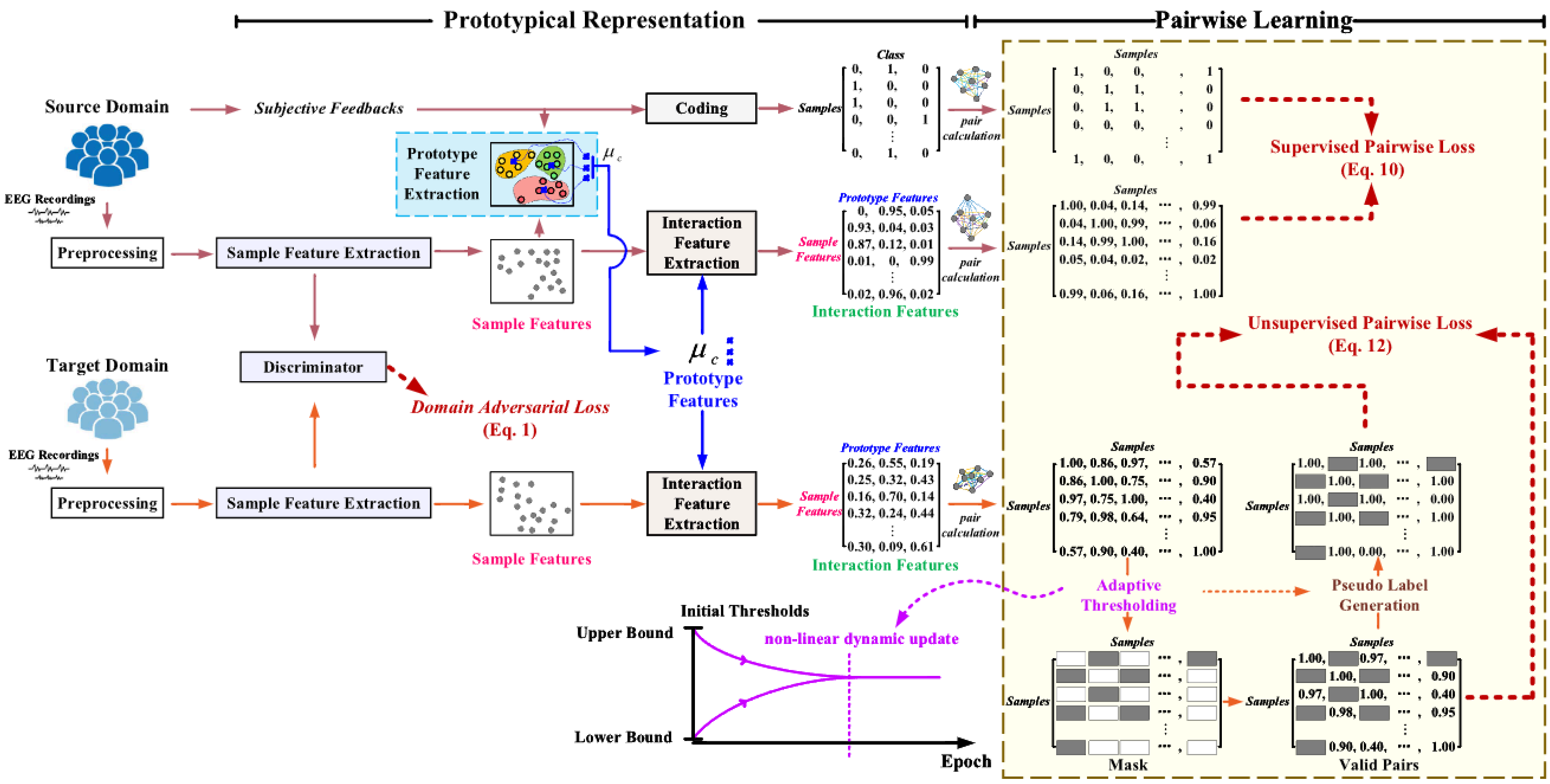

图 1. 所提出的 PR-PL 框架。在所提出的 PR-PL 中，领域对抗损失、监督成对损失和无监督成对损失分别定义于(1)、(10)和(12)。最终的目标函数是上述三种损失函数的组合，如(17)所示。

表 I
资金

Notation	Description
$\mathbb{S}\backslash\mathbb{T}$	source\target domain
$x^s\backslash x^t$	source\target feature
$y^s\backslash y^t$	source\target label
$N^s\backslash N^t$	number of source\target samples
$D^s\backslash D^t$	the source\target dataset
$f(\cdot)$	sample feature extractor
$d(\cdot)$	domain discriminator
$h(\cdot)$	bilinear operation
μ_c	prototype features
l_i	interaction features
θ_f	the parameter of the feature extractor $f(\cdot)$
θ_d	the parameter of the discriminator $d(\cdot)$
S	bilinear transformation matrix in $h(\cdot)$

将在以下成对学习中使用。在成对学习中，探索源域和目标域中的对关系。由于在模型训练过程中，目标域中的 Y 信息是未知的，因此提出了一种非监督成对学习策略，用于伪标签和掩码生成，并结合非线性动态阈值方法。

A. 样本特征提取

为了使源域和目标域数据满足独立同分布的假设并遵循相同的分布，我们基于 DANN 中引入的域对抗训练来表征样本特征。通过这种方法，缓解了源域和目标域之间的分布差异，并表征了具有域不变特性的样本特征。通过这一过程

EEG 信号中的个体差异可以得到缓解，模型的泛化能力可以得到提升[6], [24], [25], [26]。具体来说，通过采用领域对抗训练，最小化源域上样本特征 f(X)与目标域上样本特征 f(X)之间的领域差异。这里，带有参数θ的 f(·)是用于从 EEG 信号中提取样本特征的设计好的特征提取器。引入带有参数θ的领域判别器 d(·)来区分特征化样本特征（f(X)或 f(X)）是否来自源域（S）或目标域（T），其损失函数是标准的二分类交叉熵损失函数，表示为

$$L(\theta, \theta) = -\sum_{i=1}^{N} \log d(f(x)) - \sum_{i=1}^{N} \log(1 - d(f(x))), \quad (1)$$

这被称为图 1 中的域对抗损失。

在训练过程中，我们采用端到端训练方法[22]，通过引入梯度反转层实现领域对抗训练。特征提取器 f(·)最大化分类能力以提升情绪识别性能，同时判别器 d(·)最小化领域区分以减少源域和目标域之间的分布差异。领域对抗训练过程中的目标函数可以表示为

$$\min_{\theta} \max_{\theta} L(\theta) - \lambda L(\theta, \theta). \quad (2)$$

Here, $\mathcal{L}_{\text{classifier}}(\theta_f)$ is the classification loss to measure the modeling capability on the emotion recognition task. In this paper, $\mathcal{L}_{\text{classifier}}(\theta_f)$ will be realized by supervised pairwise learning on the source domain and unsupervised pairwise learning on the target domain introduced in Sections III-D and III-E below. $\mathcal{L}_{\text{disc}}(\theta_f, \theta_d)$ is the adversarial loss for the discriminator given in (1) to be trained to distinguish the sample features characterized from source and target domains. θ_f and θ_d are the parameters of $f(\cdot)$ and $d(\cdot)$. λ is a balanced hyperparameter for ensuring the stability of domain adversarial, which is given by an exponential growth method as

$$\lambda = \frac{2}{1 - \exp(-p)} - 1. \tag{3}$$

Here, p is a factor related to the training round, given by a ratio of the current training round to the maximum training round.

B. Prototype Feature Extraction

We assume that there exists a prototype representation for each emotion category. Through prototypical learning, the prototype features could be learned to indicate the representation property of every single emotion category. Based on the sample features extracted from different subjects under different emotions, we could consider these sample features are distributed around the prototype features. In other words, for each emotion category, the prototype feature could be considered the "center of mass" of all the sample features. From the perspective of a probability distribution, the prototype feature of an emotion category can be regarded as the mean value of the sample features of that emotion, and the variance of the distribution could be caused by the unstable EEG patterns, including but not limited to individual differences and the uncertainty of emotion expression. Assume that the sample features under an emotion category c obey the Gaussian distribution $N(\mu_c, \sigma_c^2)$. The prototype feature of the emotion category c could be calculated as the mean vector μ_c of the sample feature distribution. For the source domain $\{X_s, Y_s\} = \{(x_i^s, y_i^s)\}_{i=1}^{N_s}$, the corresponding sample features are characterized by the feature extractor $f(\cdot)$ defined in Section III-A, given as $f(x_i^s)$. The prototype feature vector of the emotion category c is estimated as

$$\mu_c = \frac{1}{|X_s^c|} \sum_{x_i^s \in X_s^c} f\left(x_i^s\right), \tag{4}$$

where $X_s^c = \{(x_i^s, y_i^s = c)\}_{i=1}^{|X_s^c|}$ is a collection of source domain data belonging to the emotion category c in the current mini-batch, and $|X_s^c|$ is the corresponding sample size. In order to achieve optimal prototype representation, μ_c would undergo iterative updates during the model training process. Here, μ_c could be expressed as the centroid of the sample features of X_s^c. The mean value calculation is a widely used, simple, and effective noise reduction strategy, which can make the prototype features stronger than the sample features and help to alleviate the problem of the traditional DANN network being susceptible to related noise interference [33]. It is noted that we only use the source domain here for prototype feature extraction, where the emotional label information is adopted. After the model training, the optimal prototype features of each emotion category are obtained, which would then be fixed during the model validation procedure.

C. Interaction Feature Extraction

In the traditional DANN feature extraction, the derived domain-invariant shared feature representation could be easily contaminated by the shared noises on the source and target domains. In this paper, we introduce a bilinear transformation to quantify the interactions between the sample features and the prototype features and then extract the interaction features for subsequent pairwise learning.

For a given d-dimensional sample feature $f(x_i)$ (x_i could either belong to the source domain or the target domain), its interactions with the prototype feature μ_c of the cth emotion category is measured by a bilinear transformation $h(\cdot)$, defined as

$$h\left(f\left(x_i\right), \mu_c\right) = f\left(x_i\right)^T S \mu_c, \tag{5}$$

where $S \in \mathbb{R}^{d \times d}$ is a random-initialized bilinear transformation matrix with trainable parameters and is not restricted by symmetry or positive definiteness. The bilinear transformation matrix S continues to be updated during the model learning process through a gradient backpropagation by optimizing the final objective function given in (17). If we remove the bilinear transformation matrix S from (5) and replace it with an identity matrix, then (5) could be considered as an inter product between the sample feature and the prototype feature. Compared to the inter product, a trainable transformation matrix could be more beneficial to enhance the feature representation capability for the downstream tasks, which is also evidenced in the ablation study.

Further, for a total of n emotion categories, the interaction feature of x_i is a feature vector consisting of the interactions between the corresponding sample feature $f(x_i)$ and each of the prototype features μ_c ($c = 1, \ldots, n$), specified as

$$l_i = \text{softmax}\left(\left[h\left(f\left(x_i\right), \mu_1\right), \ldots, h\left(f\left(x_i\right), \mu_n\right)\right]\right), \tag{6}$$

where $\{\mu_1, \ldots, \mu_n\}$ are the prototype features corresponding to the n emotion categories, based on the definition given in (4). To further enrich the non-linear characteristics of the interaction features and enhance the feature separability for achieving advanced emotion recognition performance, the softmax function is adopted here to strengthen the feature representation with prediction ability. Overall, l_i could be interpreted as the characterized interaction feature for mapping to a non-linear informative feature space by measuring the interaction relationship between the sample features and all the available prototype features.

D. Supervised Pairwise Learning on Source Domain

For the source domain, the traditional pointwise learning algorithms often regard the feature vector l_i^s as the predicted label of the sample feature x_i^s and use the cross-entropy loss function to match the label y_i^s of x_i^s based on supervised training. Then, if the sample features of one EEG trial match the prototype features of the cth emotion category the most, the EEG trial

这里，L(θ) 是用于衡量在情绪识别任务上建模能力的分类损失。在本文中，L(θ) 将通过在源域上进行监督成对学习以及在下面第 III-D 和 III-E 节中引入的目标域上进行无监督成对学习来实现。L(θ, θ) 是公式 (1) 中给定的判别器的对抗损失，用于训练以区分源域和目标域的特征样本。θ 和 θ 是函数 f (·) 和 d(·) 的参数。λ 是一个平衡超参数，用于确保域对抗的稳定性，它通过指数增长方法给出：

$$\lambda = 2\frac{}{1 - \exp(-p)} - 1。\quad (3)$$

这里，p 是与训练轮次相关的因素，由当前训练轮次与最大训练轮次的比值给出。

B. 原型特征提取

我们假设每个情绪类别都存在一个原型表示。通过原型学习，原型特征可以被学习用来指示每个情绪类别的表示特性。基于从不同情绪状态下不同受试者提取的样本特征，我们可以认为这些样本特征围绕原型特征分布。换句话说，对于每个情绪类别，原型特征可以被视为所有样本特征的“质心”。从概率分布的角度来看，一个情绪类别的原型特征可以被视为该情绪样本特征的均值，而分布的方差可能由不稳定的脑电图模式引起，包括但不限于个体差异和情绪表达的不确定性。假设情绪类别 c 下的样本特征服从高斯分布 N(μ, σ)，那么情绪类别 c 的原型特征可以计算为样本特征分布的均值向量μ。 对于源域 {X, Y} = {(x, y)}i，相应的样本特征由第三节 A 中定义的特征提取器 f (·) 描述，表示为 f (x)。情绪类别 c 的原型特征向量估计为

$$\mu = \frac{1}{|X|}\sum_{x\in X} f(x), \quad (4)$$

其中 X= {(x, y= c)}iis 一个属于当前小批量中情感类别 c 的源域数据集合，且|X|是对应的样本数量。为了实现最优原型表示，μ在模型训练过程中会进行迭代更新。在此，μ可以表示为 X 的样本特征的质心。均值计算是一种广泛使用、简单且有效的降噪策略，可以使原型特征比样本特征更强，并有助于缓解传统 DANN 网络易受相关噪声干扰的问题[33]。需要注意的是，我们这里仅使用源域进行原型特征提取，采用情感标签信息。模型训练后，将获得每个情感类别的最优原型特征，这些特征在模型验证过程中将保持固定。

C. 交互特征提取

在传统的 DANN 特征提取中，所得到的领域不变共享特征表示很容易受到源域和目标域上共享噪声的污染。在本文中，我们引入一个双线性变换来量化样本特征与原型特征之间的交互，然后提取交互特征以进行后续的成对学习。

对于一个给定的 d 维样本特征 f(x)（x 可以属于源域或目标域），它与第 c 类情感类别的原型特征μ的交互通过一个双线性变换 h(·)来测量，定义为

$$h(f(x), \mu) = f(x)S\mu, \quad (5)$$

其中 S ∈ R 是一个随机初始化的双线性变换矩阵，具有可训练参数，不受对称性或正定性的限制。双线性变换矩阵 S 在模型学习过程中通过梯度反向传播不断更新，以优化公式（17）中给出的最终目标函数。如果我们从（5）中移除双线性变换矩阵 S 并用单位矩阵替换它，那么（5）可以被视为样本特征与原型特征之间的交互乘积。与交互乘积相比，可训练的变换矩阵可以更有利于增强下游任务的特征表示能力，这在消融研究中也得到了证实。

此外，对于总共 n 个情绪类别，x 的交互特征是一个特征向量，由相应的样本特征 f(x) 与每个原型特征 μ(c = 1, . . ., n) 之间的交互组成，表示为

$$l = \text{softmax}\left([h(f(x), \mu), \ldots, h(f(x), \mu)]\right) \quad (6)$$

其中 {μ, . . ., μ} 是基于公式（4）定义的对应于 n 个情绪类别的原型特征。为了进一步丰富交互特征的非线性特性，增强特征可分性以实现高级情绪识别性能，这里采用了 softmax 函数来强化具有预测能力的特征表示。总体而言，l 可以解释为特征化的交互特征，通过测量样本特征与所有可用原型特征之间的交互关系，将其映射到非线性信息特征空间。

D. 源域的监督成对学习

对于源域，传统的点式学习算法通常将特征向量 l 视为样本特征 x 的预测标签，并使用交叉熵损失函数基于监督训练来匹配 x 的标签 y。然后，如果某个 EEG 试验的样本特征与第 c 个情绪类别的原型特征最匹配，该 EEG 试验

will be assigned to the cth emotion category. However, this type of pointwise learning only focuses on the relationship between sample features and prototype features and ignores the relationship among different pairs of samples. For example, the samples belonging to different emotion categories should be separated from each other, and the samples belonging to the same emotion category should be gathered together.

Considering this, we introduce pairwise learning to replace the traditional pointwise learning and conduct pair calculation to capture the inherent relationship of different pairs of samples in terms of the interaction features obtained in (6). The pairwise loss function on the source domain could be defined as

$$\mathcal{L}_{\text{class}}(\theta) = \frac{1}{N_s N_s} \sum_{i,j \in N_s} L\left(r_{ij}^s, g\left(x_i^s, x_j^s; \boldsymbol{\theta}\right)\right). \tag{7}$$

In the training process, the label information of source domain data is used to define r_{ij}^s in a supervised manner. Based on the given information of y_i^s and y_j^s, if $y_i^s = y_j^s$, then $r_{ij}^s = 1$; otherwise $r_{ij}^s = 0$. A supervised r_{ij}^s can ensure the stability of the training process and the generalization ability of the model. On the other hand, $g(x_i^s, x_j^s; \boldsymbol{\theta})$ is a similarity measurement of the samples x_i^s and x_j^s, with the parameter of θ. The loss function $L(\cdot)$ is a difference calculation of r_{ij} and $g(x_i^s, x_j^s; \boldsymbol{\theta})$, given by a two-category cross-entropy loss as

$$\begin{aligned} L\left(r_{ij}^s, g\left(x_i^s, x_j^s; \theta\right)\right) = & -r_{ij}^s \log\left(g\left(x_i^s, x_j^s; \theta\right)\right) \\ & -\left(1 - r_{ij}^s\right) \log\left(1 - g\left(x_i^s, x_j^s; \theta\right)\right), \end{aligned} \tag{8}$$

where $g(x_i^s, x_j^s; \boldsymbol{\theta})$ is a similarity measurement between x_i^s and x_j^s in terms of the characterized interaction features l_i^s and l_j^s as

$$g\left(x_i^s, x_j^s; \theta\right) = \frac{l_i^s \cdot l_j^s}{\left\|l_i^s\right\|_2 \left\|l_j^s\right\|_2}, \tag{9}$$

where $\cdot$ refers to inner product operation. A norm restriction $l/\left\|l\right\|_2$ is adopted here to make the similarity results located in the range of $[0, 1]$ and extract better and more robust feature representations for subsequent classification [34].

Overall, the objective function of the supervised pairwise learning on the source domain is defined as

$$\mathcal{L}_{\text{pairwise}}^s(\theta) = \frac{1}{N_s N_s} \sum_{i,j \in N_s} L\left(r_{ij}^s, g\left(x_i^s, x_j^s; \theta\right)\right) + \beta \mathcal{R}, \tag{10}$$

which is termed as *supervised pairwise loss* in Fig. 1. Here, $\theta = \{\theta_f, S\}$, θ_f is the parameter of feature extractor $f(\cdot)$, and S is the defined bilinear transformation matrix for interaction feature extraction. Besides, to avoid redundant feature extraction, a soft regularization $\mathcal{R}$ is introduced with a weight parameter of β, which is defined as

$$\mathcal{R} = \left\|P^T P - I\right\|_F. \tag{11}$$

Here, each row of the matrix P refers to the prototype feature belonging to one emotion category, $\|\cdot\|_F$ is a F norm of the matrix, and I is an identity matrix.

The above loss function (10) could be interpreted as a clustering loss, instead of an emotion category classification loss. The main optimization goal is to gather the EEG samples that may belong to the same emotion category and separate the EEG samples that do not belong to the same category.

E. Unsupervised Pairwise Learning on Target Domain

In this paper, we also introduce unsupervised pairwise learning to improve the feature separability on the target domain, as

$$\mathcal{L}_{\text{pairwise}}^t(\theta_f, S) = \frac{1}{N_t N_t} \sum_{i,j \in N_t} L\left(r_{ij}^t, \frac{l_i^t \cdot l_j^t}{\left\|l_i^t\right\|_2 \left\|l_j^t\right\|_2}\right), \tag{12}$$

which is termed as *unsupervised pairwise loss* in Fig. 1. Here, l_i^t and l_j^t are the interaction features of x_i^t and x_j^t on the target domain, characterized by (6). The scalar r_{ij}^t symbolizes the pairing relationship of x_i^t and x_j^t. Since the label information of the target domain is completely unknown in the training process, we cannot accurately obtain the pairing relationship as the source domain. To address this issue, we introduce an adaptive thresholding method to generate a mask for selecting valid pair relationships and producing the corresponding pseudo labels. Suppose that r_{ij}^t is defined as

$$r_{ij}^t = \begin{cases} 1, \text{ if } l_i^t \cdot l_j^t \geq \tau_h \\ 0, \text{ if } l_i^t \cdot l_j^t < \tau_l, \end{cases} \quad i, j = 1, \ldots, N_t, \tag{13}$$

where τ_h and τ_l are the upper and lower bounds to select valid pairs with high confidence for unsupervised pairwise learning (mask generation). Here, if the calculated pairwise similarity is higher or equal to the defined upper bound (τ_h), then the corresponding pseudo label of r_{ij}^t would be assigned to 1; while if the calculated pairwise similarity is lower than the defined lower bound (τ_l), then the corresponding pseudo label of r_{ij}^t would be assigned to 0. For the other pairs that do not meet the threshold requirement, we would consider that the model is uncertain about whether the sample pair is paired or not and consider these pairs as invalid results. In order to prevent incorrect optimization, the invalid pairs would be temporarily excluded from training and will not participate in the loss calculation at the current training round.

Considering the classification performance on the target domain could be quite unstable in the early training stage, we set a strict upper threshold (τ_h) and a lower threshold (τ_l) and exclude most of the pair results on the target domain to ensure the training stability. Along with the enhancement of model performance on the target domain, we can gradually lower the upper threshold and raise the lower threshold and allow more samples to participate in the training part. In other words, along with the increase in training steps, more pair relationships on the target domain could be included for model learning. Here, we form a non-linear dynamic update for adaptive thresholding, as

$$\tau_h^t = \tau_h^{t-1} - \frac{\tau_h^{t-1} - \tau_l^{t-1}}{maxepoch}, \tag{14}$$

$$\tau_l^t = \tau_l^{t-1} + \frac{\tau_h^{t-1} - \tau_l^{t-1}}{maxepoch}, \tag{15}$$

where τ_h^t and τ_l^t represent the upper and lower thresholds at the current training round t, respectively. $maxepoch$ is the

将被分配到第 c 类情感类别。然而，这种逐点学习方法只关注样本特征与原型特征之间的关系，而忽略了不同样本对之间的关系。例如，属于不同情感类别的样本应该彼此分开，而属于同一情感类别的样本应该聚集在一起。

考虑到这一点，我们引入成对学习方法来替代传统的逐点学习方法，并执行成对计算以捕获（6）中获得的交互特征的不同样本对的内在关系。源域上的成对损失函数可以定义为

$$L(\theta) = \frac{1}{NN} \sum_{i,j \in N} L\left(r, g\left(x, x; \theta\right)\right). \quad (7)$$

在训练过程中，使用源域数据的标签信息以监督方式定义 r。基于给定的 y 和 y 的信息，如果 y=y，则 r=1；否则 r=0。一个监督的 r 可以确保训练过程的稳定性以及模型的泛化能力。另一方面，$g(x, x; \theta)$是样本 x 和 x 之间的相似度度量，具有参数θ。损失函数 L(·)是 r 和 $g(x, x; \theta)$之间的差异计算，以二元交叉熵损失给出

$$L\left(r, g\left(x, x; \theta\right)\right) = -r\log\left(g\left(x, x; \theta\right)\right) - \left(1 - r\right)\log\left(1 - g\left(x, x; \theta\right)\right), \quad (8)$$

其中 $g(x, x; \theta)$是 x 和 x 在特征 l 和 l 方面的相似性度量

$$g\left(x, x; \theta\right) = \frac{l \cdot l}{\|l\| \|l\|}, \quad (9)$$

其中 · 表示内积运算，‖ ‖ 一个范数限制。‖ ‖ 这里采用 2 是为了使相似度结果位于[0, 1]范围内，并为后续分类提取更好、更鲁棒的特征表示[34]。

总体而言，源域上的监督成对学习的目标函数定义为

$$\text{Lpairwise}(\theta) = \frac{1}{NN} \sum_{i,j \in N} L\left(r, g\left(x, x; \theta\right)\right) + \beta R, \quad (10)$$

这被称为图 1 中的监督成对损失。这里，θ = {θ, S}，θ是特征提取器 f(·)的参数，S 是用于交互特征提取的预定义双线性变换矩阵。此外，为了避免冗余特征提取，引入了一个软正则化 R，其权重参数为β，定义为

$$R = \| PP - I \|_F. \quad (11)$$

这里，矩阵 P 的每一行表示属于一个情感类别的一个原型特征，‖ · ‖ 是矩阵的 F 范数，I 是单位矩阵。

上述损失函数(10)可以解释为一种聚类损失，而不是情感类别分类损失。

主要的优化目标是聚集可能属于同一情感类别的脑电样本，并分离不属于同一类别的脑电样本。

E. 目标域上的无监督成对学习

在本文中，我们还引入了无监督成对学习来提高目标域上的特征可分性，作为

$$\text{Lpairwise}(\theta, S) = \frac{1}{NN} \sum_{i,j \in N} L\left(r, \frac{l \cdot l}{\|l\| \|l\|}\right), \quad (12)$$

这被称为图 1 中的无监督成对损失。在这里，l 和 l 是 x 和 x 在目标域中的交互特征，由(6)表征。标量 r 表示 x 和 x 的配对关系。由于在训练过程中目标域的标签信息完全未知，我们无法像源域一样准确获得配对关系。为了解决这个问题，我们引入了一种自适应阈值方法来生成一个掩码，用于选择有效的配对关系并生成相应的伪标签。假设 r 被定义为

$$r = \begin{cases} 1，如果\ l \cdot l \geqslant \tau \\ 0，如果\ l \cdot l < \tau，i，j = 1，\ldots，N， \end{cases} \quad (13)$$

其中 τ 和 τ 是用于在无监督成对学习中（掩码生成）选择具有高置信度的有效对的上下界。这里，如果计算出的成对相似度高于或等于定义的上界（τ），则相应的伪标签 rw 将被赋值为 1；而如果计算出的成对相似度低于定义的下界（τ），则相应的伪标签 r 将被赋值为 0。对于不满足阈值要求的其他对，我们认为模型不确定该样本对是否成对，并将这些对视为无效结果。为了防止错误优化，无效对将被暂时从训练中排除，并且不会参与当前训练轮次的损失计算。

考虑到在早期训练阶段目标域的分类性能可能非常不稳定，我们设定了一个严格的上限阈值（τ）和一个下限阈值（τ），并排除了目标域上大部分的配对结果以确保训练稳定性。随着模型在目标域上的性能提升，我们可以逐渐降低上限阈值并提高下限阈值，允许更多样本参与训练部分。换句话说，随着训练步数的增加，目标域上的更多配对关系可以被纳入模型学习。在这里，我们形成了一个非线性动态更新机制用于自适应阈值调整，具体表示为：

$$\tau = \tau h - \frac{\tau h - \tau l}{\text{maxepoch}}, \quad (14)$$

$$\tau = \tau l + \frac{\tau h - \tau l}{\text{maxepoch}}, \quad (15)$$

其中 τ 和 τ 分别表示当前训练轮次 t 的上限阈值和下限阈值。maxepoch 是

maximum training round. Based on the given initial values τ_h^0 and τ_l^0, the calculations of τ_h^t and τ_l^t could be considered as non-linear updates with respect to the training rounds as

$$\begin{cases} \tau_h^t = \tau_h^0 - \frac{\tau_h^0 - \tau_l^0}{2} \times \left(1 - \left(1 - \frac{2}{maxepoch}\right)^t\right) \\ \tau_l^t = \tau_l^0 + \frac{\tau_h^0 - \tau_l^0}{2} \times \left(1 - \left(1 - \frac{2}{maxepoch}\right)^t\right). \end{cases} \tag{16}$$

Overall, the final objective function of the proposed PR-PL (2) could be written as a combination of the domain adversarial loss (1), the supervised pairwise loss on the source domain (10), and the unsupervised pairwise loss on the target domain (12), given as

$$\min_{\theta_f, S} \max_{\theta_d} \mathcal{L}^s_{\text{pairwise}}(\theta_f, S) + \gamma \mathcal{L}^t_{\text{pairwise}}(\theta_f, S) - \lambda \mathcal{L}_{\text{disc}}(\theta_f, \theta_d), \tag{17}$$

where γ is a hyperparameter to control the importance of the unsupervised pairwise loss on the target domain, defined as

$$\gamma = \delta \times \frac{epoch}{maxepoch}, \tag{18}$$

where $epoch$ is the current training round, and $maxepoch$ is the maximum training round. δ is a constant value. During the model training, we compute the final objective function on the basis of the data samples $(\{x^s, y^s\}, \{x^t, -\})$ from one random mini-batch with the stochastic gradient descent for parameter updating.

IV. Experimental Results

A. Emotional Databases and Data Preprocessing

We validate the proposed PR-PL on two well-known public databases: SEED [13] and SEED-IV [35]. In SEED database [13], a total of 15 subjects were invited. Each subject performed three sessions on different days and each session contained 15 trials. A total of three emotions were elicited (negative, neutral, and positive). In SEED-IV database [35], a total of 15 subjects participated in the experiment. For each subject, a total of 3 sessions were performed on different days and each session contained 24 trials. A total of four emotions were elicited (happiness, sadness, fear, and neutral). The EEG signals of both SEED and SEED-IV databases were simultaneously collected using the 62-channel ESI Neuroscan system.

For EEG preprocessing, the data sampling rate was first downsampled to 200 Hz, and the contaminated noises (e.g., EMG and EOG) were manually removed. Then, the data were filtered by a band-pass filter of 0.3 Hz to 50 Hz. For each trial, the data was divided into a number of segments with a length of 1 s. Based on the pre-defined five frequency bands: Delta (1-3 Hz), Theta (4-7 Hz), Alpha (8-13 Hz), Beta (14-30 Hz), and Gamma (31-50 Hz), the corresponding differential entropy (DE) features were extracted to represent the logarithm energy spectrum in a specific frequency band and total 310 features (5 frequency bands $\times$ 62 channels) were obtained for one EEG segment. Then, all the features in a trial were smoothed with the linear dynamic system (LDS) method, which can utilize the time dependency of emotion changes and filter out emotion unrelated and noisy EEG components [36]. During the cross-validation, there would not exist any information leakage problem, since the source domain data and the target domain data are obtained from distinct trials. Here, the EEG data preprocessing adheres to the same standards as the prior studies for a fair comparison with the literature.

B. Implementation Results

In our experiments, the feature extractor $f(\cdot)$ and the discriminator $d(\cdot)$ are both made up of the multilayer perceptron (MLP) with the Relu activation function. All the parameters are randomly initialized from a uniform distribution. The bilinear operator matrix S given in (5) is also randomly initialized, with the size of 64×64. In the model architecture, the feature extractor structure is designed as 310 (input layer)-64 (hidden layer 1)-Relu activation-64 (hidden layer 2)-Relu activation-64 (output feature layer). The discriminator structure is designed as 64 (input layer)-64 (hidden layer 1)-Relu activation-dropout layer-64 (hidden layer 2)-1 (output layer)-Sigmoid activation. Besides, we adopt an RMSprop optimizer for network training, which shows a better performance than the other classic optimizers. The learning rate is set to 1e-3 and the mini-batch size for training is 96. To avoid overfitting problems, we use $L2$ regularization with the weight of 1e-5 in the network. The β value in (10) is assigned as 0.01, and the δ value in (18) is set to 2. The threshold τ_h^0 and τ_l^0 are initialized as 0.9 and 0.5 respectively. All the models are trained on an NVIDIA GeForce RTX 2080 GPU, with CUDA 10.0 using the Pytorch API. It is notably that, in the model training process, only the raw target domain data are used without any label information involved, which is the same as the previous EEG-based emotion recognition methods using transfer learning framework [5], [8], [37], [38].

C. Experiment Protocols

To make a thorough comparison with the literature, we validate the proposed PR-PL using four different cross-validation evaluation protocols under consideration of the variants in subjects and sessions. *(1) Within-subject cross-session cross-validation.* For one subject's data, the first two sessions are used as the source, and the latter session is used as the target. The average accuracies and standard deviations across subjects are calculated as the final results. *(2) Within-subject single-session cross-validation.* Following the cross-validation evaluation protocol presented in the existing studies [13], [35], for each session of one subject, we use the first 9 (SEED) or 16 (SEED-IV) trials as the source and the rest 6 (SEED) or 8 (SEED-IV) trials as the target. The results are reported as the average performance across all the subjects. The above two cross-validation evaluation protocols are subject-dependent to estimate the model stability within one subject. To further evaluate the robustness and stability of the proposed PR-PL across different subjects, we also perform the model validation on the following two evaluation protocols (subject-independent). *(3) Cross-subject cross-session leave-one-subject-out cross-validation.* To estimate the model efficacy on the unknown subject(s) and session(s), we use one

最大训练轮次。基于给定的初始值τ和τ，τ和τ的计算可被视为相对于训练轮次的非线性更新，即

$$\begin{cases} \tau = \tau - 2 \times \left(1 - \left(1 - \frac{2}{\text{maxepoch}}\right)\right) \\ \tau = \tau + 2 \times \left(1 - \left(1 - \frac{2}{\text{最大迭代次数}}\right)\right) \end{cases} \quad (16)$$

总体而言，所提出的 PR-PL（2）的最终目标函数可以表示为领域对抗损失（1）、源域上的监督成对损失（10）和目标域上的无监督成对损失（12）的组合，表示为

$$\min_{\theta,S}\max_{\theta} \text{Lpairwise}(\theta, S)+\gamma\text{Lpairwise}(\theta, S)-\lambda L(\theta, \theta),$$

(17) 其中 γ 是一个超参数，用于控制无监督成对损失在目标域中的重要性，定义为

$$\gamma = \delta \times \frac{\text{epoch}}{\text{maxepoch}}, \quad (18)$$

其中，epoch 表示当前训练轮次，maxepoch 表示最大训练轮次。δ是一个常数值。在模型训练过程中，我们基于一个随机的小批量数据样本({x, y}, {x, −})，使用随机梯度下降法来计算最终的目标函数，并更新参数。

IV. ER

A. 情感数据库与数据预处理

我们在两个著名的公开数据库上验证了所提出的 PR-PL：SEED [13] 和 SEED-IV [35]。在 SEED 数据库 [13] 中，共邀请了 15 名受试者。每位受试者在不同天数进行了三次测试，每次测试包含 15 次试验。总共诱发了三种情绪（负面、中性、正面）。在 SEED-IV 数据库 [35] 中，共 15 名受试者参与了实验。每位受试者在不同天数进行了三次测试，每次测试包含 24 次试验。总共诱发了四种情绪（快乐、悲伤、恐惧、中性）。SEED 和 SEED-IV 数据库的脑电信号均使用 62 通道的 ESI Neuroscan 系统同时采集。

在脑电图（EEG）预处理中，首先将数据采样率下采样至 200 Hz，并手动去除受污染的噪声（例如肌电图和眼电图）。然后，使用 0.3 Hz 至 50 Hz 的带通滤波器对数据进行滤波。对于每个试验，将数据划分为多个长度为 1 秒的片段。基于预定义的五个频段：Delta（1-3 Hz）、Theta（4-7 Hz）、Alpha（8-13 Hz）、Beta（14-30 Hz）和 Gamma（31-50 Hz），提取相应的差分熵（DE）特征来表示特定频段的对数能量频谱。对于每个 EEG 片段，获得 310 个特征（5 个频段×62 个通道）。然后，使用线性动态系统（LDS）方法对所有试验中的特征进行平滑处理。该框架能够利用情绪变化的时间依赖性，过滤掉与情绪无关和含噪声的 EEG 成分[36]。在交叉验证过程中，不会存在信息泄露问题，因为源域数据和目标域数据来自不同的试验。在此，EEG 数据预处理遵循先前研究的标准，以便与文献进行公平比较。

B. 实施结果

在我们的实验中，特征提取器 f (·) 和判别器 d(·) 都由具有 ReLU 激活函数的多层感知器 (MLP) 组成。所有参数都从均匀分布中随机初始化。公式 (5) 中给出的双线性算子矩阵 S 也随机初始化，大小为 64 × 64。在模型架构中，特征提取器结构设计为 310（输入层）-64（隐藏层 1）-ReLU 激活-64（隐藏层 2）-ReLU 激活-64（输出特征层）。判别器结构设计为 64（输入层）-64（隐藏层 1）-ReLU 激活-dropout 层-64（隐藏层 2）-1（输出层）-Sigmoid 激活。此外，我们采用 RMSprop 优化器进行网络训练，其表现优于其他经典优化器。学习率设置为 1e-3，训练的 mini-batch 大小为 96。为了避免过拟合问题，我们在网络中使用权重为 1e-5 的 L2 正则化。公式 (10) 中的 β 值设为 0.01，公式 (18) 中的 δ 值设为 2。阈值 τ 和 τ 分别初始化为 0.9 和 0.5。所有模型均在 NVIDIA GeForce RTX 2080 GPU 上使用 CUDA 10.0 和 Pytorch API 进行训练。值得注意的是，在模型训练过程中，仅使用了原始目标域数据，未涉及任何标签信息，这与之前使用迁移学习框架的基于脑电信号的情绪识别方法[5]、[8]、[37]、[38]相同。

C. 实验协议

为了与文献进行彻底比较，我们在考虑受试者和会话差异的情况下，使用四种不同的交叉验证评估协议验证所提出的 PR-PL 方法。(1) 受试者内跨会话交叉验证。对于一位受试者的数据，前两个会话被用作源数据，后一个会话被用作目标数据。计算所有受试者的平均准确率和标准差作为最终结果。(2) 受试者内单会话交叉验证。遵循现有研究中[13]、[35]提出的交叉验证评估协议，对于一位受试者的每个会话，我们使用前 9（SEED）或 16（SEED-IV）次试验作为源数据，其余 6（SEED）或 8（SEED-IV）次试验作为目标数据。结果以所有受试者的平均性能报告。上述两种交叉验证评估协议是受试者相关的，用于评估单个受试者内的模型稳定性。为了进一步评估所提出的 PR-PL 在不同受试者中的鲁棒性和稳定性，我们还对以下两个评估协议（受试者无关）进行了模型验证。 (3) 跨被试跨会话留一被试交叉验证。为了评估模型对未知被试和会话的有效性，我们使用一种

TABLE II
MODEL PERFORMANCE ON SEED USING WITHIN-SUBJECT CROSS-SESSION CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
RF* [40]	76.42±11.15	KNN* [41]	75.68±13.82
TCA* [20]	74.27±12.88	CORAL* [42]	84.18±09.81
SA* [43]	69.84±09.46	GFK* [44]	78.79±09.39
Deep learning methods			
DAN* [45]	89.16±07.90	SimNet* [33]	86.88±07.83
DDC* [46]	91.14±05.61	ADA [27]	89.13±07.13
DANN* [22]	89.45±06.74	MMD [27]	84.38±12.05
JDA-Net [27]	91.17±08.11	DCORAL* [47]	88.67±06.25
PR-PL			**93.18±06.55**

TABLE III
MODEL PERFORMANCE ON SEED-IV USING WITHIN-SUBJECT CROSS-SESSION CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
RF* [40]	60.27±16.36	KNN* [41]	54.18±16.28
TCA* [20]	51.88±15.84	CORAL* [42]	66.06±15.13
SA* [43]	52.81±09.53	GFK* [44]	56.14±12.15
Deep learning methods			
DCORAL [48]	65.10±13.20	DAN [48]	60.20±10.20
DDC [48]	68.80±16.60	MEERNet [48]	72.10±14.10
PR-PL			**74.62±14.15**

subject's all sessions' data as the target and the remaining subjects' all sessions as the source. Take the SEED database as an example. Based on a total of 3394 samples in a single session for each subject, the data size of the source domain in the cross-subject cross-session is 14 subjects $\times$ 3 sessions $\times$ 3394 samples, and that of the target domain is 1 subject $\times$ 3 sessions $\times$ 3394 samples. We repeat the training validation until each subject is treated as the target once and report the average performance across all the subjects as the final results. *(4) Cross-subject single-session leave-one-subject-out cross-validation.* It is the most widely used validation scheme in EEG-based emotion recognition tasks [5], [26], [27], [39]. One subject's single session's data is treated as the target and the other remaining subjects' single session's data is used as the source. Take the SEED database as an example. Based on a total of 3,394 samples in a single session for each subject, the data size of the source domain in the cross-subject single-session is 14 subjects $\times$ 1 session $\times$ 3,394 samples, and that of the target domain is 1 subject $\times$ 1 session $\times$ 3,394 samples. We repeat the training validation until each subject is treated as the target once and report the average performance across all the subjects as the final results. Here, same as the other existing studies, we only consider the first session in the cross-subject single-session cross-validation.

In the following, the model performance under the aforementioned four different cross-validation evaluation protocols is reported and compared with the existing literature. Here, the model results reproduced by us are indicated by *.

D. Within-Subject Cross-Session Cross-Validation Results

We evaluate the model performance on each subject's data across all sessions. The final experimental results are the average accuracies and standard deviations across subjects, as shown in Table II (SEED) and Table III (SEED-IV). Compared to the experimental results presented in the existing traditional machine learning methods and deep learning methods, the proposed PR-PL achieves superior performance in both SEED and SEED-IV databases, where the results are 93.18%±6.55% for SEED (three-class emotion recognition) and 74.62%±14.15% for SEED-IV (four-class emotion recognition).

TABLE IV
MODEL PERFORMANCE ON SEED USING WITHIN-SUBJECT SINGLE-SESSION CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
SVM [50]	83.99±09.72	GRSLR [52]	87.39±08.64
RF [50]	78.46±11.77	GSCCA [53]	82.96±09.95
CCA [50]	77.63±13.21	DBN [13]	86.08±08.34
Deep learning methods			
DGCNN [51]	90.40±08.49	RGNN [8]	94.24±05.95
ATDD-DANN [25]	91.08±06.43	BiHDM [50]	93.12±06.06
R2G-STNN [6]	93.38±05.96	SimNet* [33]	90.13±10.84
BiDANN [24]	92.38±07.04	STRNN [54]	89.50±07.63
GCNN [50]	87.40±09.20	DANN [50]	91.36±08.30
PR-PL			**94.84±09.16**

TABLE V
MODEL PERFORMANCE ON SEED-IV USING WITHIN-SUBJECT SINGLE-SESSION CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
SVM [50]	56.61±20.05	GRSLR [52]	69.32±19.57
RF [50]	50.97±16.22	GSCCA [53]	69.08±16.66
CCA [50]	54.47±18.48	DBN [13]	66.77±07.38
Deep learning methods			
DGCNN [51]	69.88±16.29	RGNN [8]	79.37±10.54
GCNN [50]	68.34±15.42	BiHDM [50]	74.35±14.09
A-LSTM [50]	69.50±15.45	SimNet* [33]	71.38±13.12
BiDANN [24]	70.29±12.63	DANN [50]	63.07±12.66
PR-PL			**83.33±10.61**

E. Within-Subject Single-Session Cross-Validation Results

To evaluate the model effectiveness on the data that are collected from the same subject on the same day (session), we conduct within-subject single-session cross-validation as the previous studies [6], [8], [24], [49], [50]. Tables IV and V report the corresponding experimental results on SEED and SEED-IV databases, respectively. Compared to the existing methods (such as BiDANN [24], DGCNN [49], BiHDM [50], RGNN [8]), the proposed PR-PL achieves promising results on both two benchmark databases (SEED: 94.84±09.16; SEED-IV: 83.33±10.61).

表 II
MPSEED UW -SC -S
C -V

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
RF* [40]	76.42±11.15	KNN* [41]	75.68±13.82
TCA* [20]	74.27±12.88	CORAL* [42]	84.18±09.81
SA* [43]	69.84±09.46	GFK* [44]	78.79±09.39
Deep learning methods			
DAN* [45]	89.16±07.90	SimNet* [33]	86.88±07.83
DDC* [46]	91.14±05.61	ADA [27]	89.13±07.13
DANN* [22]	89.45±06.74	MMD [27]	84.38±12.05
JDA-Net [27]	91.17±08.11	DCORAL* [47]	88.67±06.25
PR-PL			**93.18±06.55**

将所有受试者的所有会话数据作为目标，其余受试者的所有会话数据作为来源。以 SEED 数据库为例。基于每个受试者在单个会话中总共 3394 个样本，跨受试者跨会话的源域数据大小为 14 受试者×3 会话×3394 个样本，目标域数据大小为 1 受试者×3 会话×3394 个样本。我们重复训练验证，直到每个受试者都作为目标被处理一次，并报告所有受试者的平均性能作为最终结果。（4）跨受试者单会话留一受试者交叉验证。这是基于脑电图的情绪识别任务中最广泛使用的验证方案[5]、[26]、[27]、[39]。一个受试者的单个会话数据作为目标，其余受试者的单个会话数据作为来源。以 SEED 数据库为例。基于每个受试者在单个会话中总共 3394 个样本，跨受试者单会话的源域数据大小为 14 受试者×1 会话×3394 个样本，目标域数据大小为 1 受试者×1 会话×3394 个样本。 我们重复进行训练和验证，直到每个受试者都作为目标被处理一次，并报告所有受试者的平均性能作为最终结果。在这里，与其他现有研究一样，我们仅考虑跨受试者单会话交叉验证中的第一个会话。

在下文中，报告并与其他现有文献比较了模型在上述四种不同的交叉验证评估协议下的性能。这里，我们复现的模型结果用*表示。

表 III
MPSEED-IV UW -SC -S
C -V

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
RF* [40]	60.27±16.36	KNN* [41]	54.18±16.28
TCA* [20]	51.88±15.84	CORAL* [42]	66.06±15.13
SA* [43]	52.81±09.53	GFK* [44]	56.14±12.15
Deep learning methods			
DCORAL [48]	65.10±13.20	DAN [48]	60.20±10.20
DDC [48]	68.80±16.60	MEERNet [48]	72.10±14.10
PR-PL			**74.62±14.15**

表 IV
MPSEED UW -SS-S
C -V

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
SVM [50]	83.99±09.72	GRSLR [52]	87.39±08.64
RF [50]	78.46±11.77	GSCCA [53]	82.96±09.95
CCA [50]	77.63±13.21	DBN [13]	86.08±08.34
Deep learning methods			
DGCNN [51]	90.40±08.49	RGNN [8]	94.24±05.95
ATDD-DANN [25]	91.08±06.43	BiHDM [50]	93.12±06.06
R2G-STNN [6]	93.38±05.96	SimNet* [33]	90.13±10.84
BiDANN [24]	92.38±07.04	STRNN [54]	89.50±07.63
GCNN [50]	87.40±09.20	DANN [50]	91.36±08.30
PR-PL			**94.84±09.16**

表 V
MPSEED-IV UW -S
S-SC -V

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
SVM [50]	56.61±20.05	GRSLR [52]	69.32±19.57
RF [50]	50.97±16.22	GSCCA [53]	69.08±16.66
CCA [50]	54.47±18.48	DBN [13]	66.77±07.38
Deep learning methods			
DGCNN [51]	69.88±16.29	RGNN [8]	79.37±10.54
GCNN [50]	68.34±15.42	BiHDM [50]	74.35±14.09
A-LSTM [50]	69.50±15.45	SimNet* [33]	71.38±13.12
BiDANN [24]	70.29±12.63	DANN [50]	63.07±12.66
PR-PL			**83.33±10.61**

D. 受试者内跨会话交叉验证结果

我们在每个受试者的所有会话数据上评估模型性能。最终实验结果为受试者间的平均准确率和标准差，如表 II（SEED）和表 III（SEED-IV）所示。与现有传统机器学习方法及深度学习方法报告的实验结果相比，提出的 PR-PL 在 SEED 和 SEED-IV 数据库中均实现了更优性能，其中 SEED 数据库（三分类情绪识别）的结果为 93.18%±6.55%，SEED-IV 数据库（四分类情绪识别）的结果为 74.62%±14.15%。

E. 基于同一受试者单次会话的交叉验证结果

为了评估模型在从同一受试者同一天（会话）收集的数据上的有效性，我们进行了基于同一受试者单次会话的交叉验证，如先前研究[6]、[8]、[24]、[49]、[50]所述。表 IV 和表 V 分别报告了在 SEED 和 SEED-IV 数据库上的相应实验结果。与现有方法（如 BiDANN[24]、DGCNN[49]、BiHDM[50]、RGNN[8]）相比，所提出的 PR-PL 在两个基准数据库（SEED: 94.84±09.16；SEED-IV: 83.33±10.61）上均取得了令人满意的结果。

TABLE VI
MODEL PERFORMANCE ON SEED USING CROSS-SUBJECT CROSS-SESSION LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
RF* [40]	69.60±07.64	KNN* [41]	60.66±07.93
SVM* [55]	68.15±07.38	Adaboost* [56]	71.87±05.70
TCA* [20]	64.02±07.96	CORAL* [42]	68.15±07.83
SA* [43]	61.41±09.75	GFK* [44]	66.02±07.59
Deep learning methods			
DCORAL* [47]	81.97±05.16	DAN* [45]	81.04±05.32
DDC* [46]	82.17±04.96	DANN* [22]	81.08±05.88
PR-PL			**85.56±04.78**

TABLE VII
MODEL PERFORMANCE ON SEED-IV USING CROSS-SUBJECT CROSS-SESSION LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
RF* [40]	50.98±09.20	KNN* [41]	40.83±07.28
SVM [57]	51.78±12.85	Adaboost* [56]	53.44±09.12
TCA [50]	56.56±13.77	CORAL* [42]	49.44±09.09
SA [50]	64.44±09.46	GFK* [44]	45.89±08.27
KPCA [57]	51.76±12.89	DNN [57]	49.35±09.74
Deep learning methods			
DGCNN [49]	52.82±09.23	DAN [57]	58.87±08.13
RGNN [8]	73.84±08.02	BiHDM [50]	69.03±08.66
BiDANN [24]	65.59±10.39	DANN [57]	54.63±08.03
PR-PL			**74.92±07.92**

TABLE VIII
MODEL PERFORMANCE ON SEED USING CROSS-SUBJECT SINGLE-SESSION LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
TKL [24]	63.54±15.47	T-SVM [24]	72.53±14.00
TCA [50]	63.64±14.88	TPT [57]	75.17±12.83
KPCA [57]	61.28±14.62	GFK [50]	71.31±14.09
SA [50]	69.00±10.89	DICA [58]	69.40±07.80
DNN [57]	61.01±12.38	SVM [57]	58.18±13.85
Deep learning methods			
DGCNN [49]	79.95±09.02	DAN [57]	83.81±08.56
RGNN [8]	85.30±06.72	BiHDM [50]	85.40±07.53
WGAN-GP [26]	87.10±07.10	MMD [27]	80.88±10.10
ATDD-DANN [25]	90.92±01.05	JDA-Net [27]	88.28±11.44
R2G-STNN [6]	84.16±07.63	SimNet* [33]	81.58±05.11
BiDANN [24]	83.28±09.60	DResNet [58]	85.30±08.00
ADA [27]	84.47±10.65	DANN [27]	81.65±09.92
PR-PL			**93.06±05.12**

TABLE IX
MODEL PERFORMANCE ON SEED-IV USING CROSS-SUBJECT SINGLE-SESSION LEAVE-ONE-SUBJECT-OUT CROSS-VALIDATION

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
Traditional machine learning methods			
GFHF [59]	49.29±07.60	GFK [59]	44.04±09.31
TCA [59]	37.01±10.47	JDA [59]	54.57±05.47
GAKT [59]	58.49±06.23	MIDA [59]	60.22±08.69
JAGP [59]	78.54±07.33	MS-STM [5]	61.41±09.72
Deep learning methods			
DGCNN [49]	68.73±08.34	JDA-Net [27]	70.83±10.25
UDDA [60]	73.14±09.43	MS-MDA [61]	59.34±05.48
DANN [60]	54.63±08.03	DAN [60]	58.87±08.13
WGAN-GP [26]	60.60±15.76	MWACN [62]	74.60±10.77
PR-PL			**81.32±08.53**

F. Cross-Subject Cross-Session Leave-One-Subject-Out Cross-Validation Results

Considering the possible real-world application scenarios in aBCI, we further validate the model performance when the data are from different subjects and collected on different days (sessions). As reported in Tables VI and VII, the emotion recognition performance of the proposed PR-PL is 85.56%±4.78% for three-class classification task on SEED and 74.92%±7.92% for four-class classification task on SEED-IV. Due to the variants in subjects and sessions, this evaluation protocol poses the greatest challenge to the model's effectiveness in the EEG-based emotion recognition tasks among the four different cross-validation evaluation protocols (within-subject cross-session: variant in session only; within-subject single-session: n/a; cross-subject cross-session: variants in both subjects and sessions; cross-subject single-session: variants in subjects only). Compared to the existing studies, the proposed PR-PL increases the classification accuracy to 3.39% and 1.08% for SEED and SEED-IV, with smaller standard deviations. These results demonstrate the proposed PR-PL has good effectiveness and generalizability in addressing the variations among subjects and sessions for the EEG-based emotion recognition tasks.

G. Cross-Subject Single-Session Leave-One-Subject-Out Cross-Validation Results

The cross-subject single-session leave-one-subject-out cross-validation is the most popular evaluation protocol in current aBCI systems, which provides an objective measurement to evaluate the model stability when different subjects' data are considered. In Table VIII, we summarize the experimental results on the SEED database and compare them with the literature. The results show the proposed PR-PL achieves the best performance (93.06±05.12), which leads 2.14% against the reported results in the literature. Compared to the latest proposed DANN-based deep transfer learning networks in which both source and target domain data are used for modeling (e.g., ATDD-DANN [25], R2G-STNN [6], BiHDM [50], BiDANN [24], and WGAN-GP [26]), the proposed PR-PL with pairwise learning can avoid the inherent defects of DANN design (e.g., only considers feature separability on source domain). Similar observations are also found in the SEED-IV results (Table IX), where the proposed PR-PL achieves superior performance with 6.72% enhancement.

表 VI
MPSEED UC -SC -S
L-O-S-OC -V

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
	Traditional machine learning methods		
RF* [40]	69.60±07.64	KNN* [41]	60.66±07.93
SVM* [55]	68.15±07.38	Adaboost* [56]	71.87±05.70
TCA* [20]	64.02±07.96	CORAL* [42]	68.15±07.83
SA* [43]	61.41±09.75	GFK* [44]	66.02±07.59
	Deep learning methods		
DCORAL* [47]	81.97±05.16	DAN* [45]	81.04±05.32
DDC* [46]	82.17±04.96	DANN* [22]	81.08±05.88
PR-PL			**85.56±04.78**

表 VII
MPSEED-IV UC -SC -S
L-O-S-OC -V

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
	Traditional machine learning methods		
RF* [40]	50.98±09.20	KNN* [41]	40.83±07.28
SVM [57]	51.78±12.85	Adaboost* [56]	53.44±09.12
TCA [50]	56.56±13.77	CORAL* [42]	49.44±09.09
SA [50]	64.44±09.46	GFK* [44]	45.89±08.27
KPCA [57]	51.76±12.89	DNN [57]	49.35±09.74
	Deep learning methods		
DGCNN [49]	52.82±09.23	DAN [57]	58.87±08.13
RGNN [8]	73.84±08.02	BiHDM [50]	69.03±08.66
BiDANN [24]	65.59±10.39	DANN [57]	54.63±08.03
PR-PL			**74.92±07.92**

表 VIII
MPSEED UC -SS-S
L-O-S-OC -V

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
	Traditional machine learning methods		
TKL [24]	63.54±15.47	T-SVM [24]	72.53±14.00
TCA [50]	63.64±14.88	TPT [57]	75.17±12.83
KPCA [57]	61.28±14.62	GFK [50]	71.31±14.09
SA [50]	69.00±10.89	DICA [58]	69.40±07.80
DNN [57]	61.01±12.38	SVM [57]	58.18±13.85
	Deep learning methods		
DGCNN [49]	79.95±09.02	DAN [57]	83.81±08.56
RGNN [8]	85.30±06.72	BiHDM [50]	85.40±07.53
WGAN-GP [26]	87.10±07.10	MMD [27]	80.88±10.10
ATDD-DANN [25]	90.92±01.05	JDA-Net [27]	88.28±11.44
R2G-STNN [6]	84.16±07.63	SimNet* [33]	81.58±05.11
BiDANN [24]	83.28±09.60	DResNet [58]	85.30±08.00
ADA [27]	84.47±10.65	DANN [27]	81.65±09.92
PR-PL			**93.06±05.12**

表 IX
MPSEED-IV UC -SS-S
L-O-S-OC -V

Methods	$P_{acc}(\%)$	Methods	$P_{acc}(\%)$
	Traditional machine learning methods		
GFHF [59]	49.29±07.60	GFK [59]	44.04±09.31
TCA [59]	37.01±10.47	JDA [59]	54.57±05.47
GAKT [59]	58.49±06.23	MIDA [59]	60.22±08.69
JAGP [59]	78.54±07.33	MS-STM [5]	61.41±09.72
	Deep learning methods		
DGCNN [49]	68.73±08.34	JDA-Net [27]	70.83±10.25
UDDA [60]	73.14±09.43	MS-MDA [61]	59.34±05.48
DANN [60]	54.63±08.03	DAN [60]	58.87±08.13
WGAN-GP [26]	60.60±15.76	MWACN [62]	74.60±10.77
PR-PL			**81.32±08.53**

F. 跨被试跨会话留一被试交叉验证结果

考虑到在 aBCI 中的实际应用场景，我们进一步验证了当数据来自不同被试且在不同天（会话）收集时模型性能。如表 VI 和表 VII 所示，所提出的 PR-PL 在 SEED 上的三分类任务情绪识别性能为 85.56%±4.78%，在 SEED-IV 上的四分类任务情绪识别性能为 74.92%±7.92%。由于被试和会话的变化，该评估协议在四种不同的交叉验证评估协议（被试内跨会话：仅会话变化；被试内单会话：不适用；跨被试跨会话：被试和会话均变化；跨被试单会话：仅被试变化）中对模型在基于脑电的情绪识别任务中的有效性构成了最大挑战。与现有研究相比，所提出的 PR-PL 将分类准确率提高了 3.39%（SEED）和 1.08%（SEED-IV），且标准差更小。 这些结果展示了所提出的 PR-PL 在处理基于脑电信号的情绪识别任务中，对于被试和会话之间的变化具有良好的有效性和泛化能力。

G. 跨被试单会话留一被试交叉验证结果

跨被试单会话留一被试交叉验证是目前 aBCI 系统中最流行的评估协议，它提供了客观的测量方法来评估模型稳定性，尤其是在考虑不同被试数据时。在表 VIII 中，我们总结了在 SEED 数据库上的实验结果，并与文献中的结果进行了比较。结果表明，所提出的 PR-PL 达到了最佳性能（93.06±05.12），比文献中报道的结果领先 2.14%。与最新提出的基于 DANN 的深度迁移学习网络（其中源域和目标域数据都用于建模，例如 ATDD-DANN [25]、R2G-STNN [6]、BiHDM [50]、BiDANN [24]和 WGANGP [26]）相比，所提出的 PR-PL 通过成对学习可以避免 DANN 设计的固有缺陷（例如，仅考虑源域上的特征可分性）。在 SEED-IV 结果（表 IX）中也发现了类似的观察结果，其中所提出的 PR-PL 实现了 6.72%的性能提升。

TABLE X
THE ABLATION STUDY OF OUR PROPOSED MODEL

Ablation study about training strategy	P_{acc}
w/o discriminator and target information	83.30±04.21
w/o pairwise learning on the source and target	87.50±06.64
w/o pairwise learning on the target	88.81±06.63
w/o the bilinear transformation matrix S	88.31±04.42
w/o prototypical representation	91.00±04.65
w/o pseudo labeling method	92.13±05.90
About hyperparameter controlling strategy	P_{acc}
w/ fixed pseudo labeling	89.92±07.21
w/ linear dynamic pseudo labeling	92.46±04.95
w/ fixed γ for unsupervised pairwise loss in Eq. 17	89.47±10.22
PR-PL	**93.06±05.12**

V. DISCUSSION

We further evaluate the effect on model performance when different parameter settings are adopted. All the present results in this session are based on the SEED database using the commonly used cross-subject single-session leave-one-subject-out cross-validation evaluation protocol in the EEG-based emotion recognition tasks.

A. Ablation Study

We conduct the ablation study to systematically explore the effectiveness of different components in the proposed model and examine the corresponding contributions to the overall performance. As shown in Table X, it is found that the introduction of domain adversarial training can greatly enhance the emotion recognition performance on the target domain. When the model is without discriminator and target domain information, the recognition accuracy reduces from 93.06% to 83.30%. Such a significant drop shows the significant impact of individual differences problem on model performance and highlights the great potential of transfer learning in aBCI applications. Besides, the results show a combination of pairwise learning on the source and target domain benefits to the model performance, where the recognition accuracy is increased by 6.35% (from 87.50% to 93.06%). Without a consideration of the feature separability on the target domain (w/o pairwuse learning on the target), the model performance would drop from 93.06% to 88.81%. In addition, the introduction of the bilinear transformation matrix S in (5) can improve the model performance, where the recognition accuracy could be increased by 4.75% (from 88.31% to 93.06%). In a comparison between with and without pseudo labeling methods, incrementally including the valid pairs from the target domain in the training process is helpful to the learning stability, as shown by the performance rise from 92.13% to 93.06%. Further, the modeling effectiveness with different pseudo labeling methods is examined. The corresponding accuracy increases from 89.92% to 92.46% when the pseudo labeling method changes from fixed to linear dynamic. The accuracy further increases to 93.06%, when a nonlinear dynamic-based adaptive pseudo labeling method is adopted. The results show a non-linear dynamic-based adaptive pseudo labeling could be helpful to screen out the valid paired samples and improve the model trainability. For the final loss function given in (17), instead of using a fixed γ for the unsupervised pairwise loss on the target domain, we propose to update γ gradually along with the training epochs to prevent model learning failures in the early training stage and balance the relationships among different losses. The benefit of a dynamic γ is also reflected in the ablation study, where the recognition accuracy increases from 89.47% and 93.06%. All the reported results using different components in the proposed PR-PL exist significant differences, with a p-value smaller than 0.001.

B. Effect of Noisy Labels

To further verify the model robustness during noisy label learning, we randomly contaminate the source labels with η% noises and test the corresponding model performance on unknown target data. Specifically, we replace η% real labels in Y_s using randomly generated labels and train the model in a supervised learning manner. Then, we test the trained model performance on the target domain. Note here that the noisy contamination is only conducted on the source domain, as the target domain needs to be used for model evaluation. In the implementation, η% value is adjusted to 10%, 20%, and 30%, respectively. The corresponding model accuracies with the standard deviations are 89.22%±6.05%, 88.39%±6.73%, and 87.71%±5.02%. It shows that, with an increase of label noise ratio from 10% to 30%, the model performance decreases slightly, with a decrease rate of 1.69%. These results demonstrate the proposed PR-PL is a reliable model which has a better tolerance to noisy labels. In the future works, the recently proposed novel methods, such as [62] and [63], could be incorporated to further eliminate more general noises in EEG signals and improve the model stability in the cross-corpus applications.

C. Confusion Matrices

To qualitatively study the model performance in each emotion class, we visually analyze the confusion matrices and compare results with the latest models [8], [24], [50]. As shown in Fig. 2, it shows all the models are good at distinguishing positive emotions from other emotions (the recognition rates are all above 90%), but it is relatively poor at distinguishing negative and neutral emotions. For example, the recognition rate of neutral in BiDANN [24] is even lower than 80% (76.72%). Compared to the existing methods ((a), (b), and (c)), our proposed model can enhance the model recognition ability, especially for distinguishing neutral and negative emotions. As shown in (d), the recognition rates for negative, neutral, and positive emotions are 92.10%, 90.39%, and 96.50%. More confusion matrix results corresponding to the other cross-validation evaluation protocols such as within-subject cross-session, within-subject single-session, cross-subject cross-session leave-one-subject-out are reported in Appendix I of Supplementary Materials, available online.

D. Visualization of Learned Representation

To verify the effectiveness of the proposed model from a more intuitive perspective, we visualize the characterized sample and interaction features of source and target domains using

表 X
TASOPM

Ablation study about training strategy	P_{acc}
w/o discriminator and target information	83.30±04.21
w/o pairwise learning on the source and target	87.50±06.64
w/o pairwise learning on the target	88.81±06.63
w/o the bilinear transformation matrix S	88.31±04.42
w/o prototypical representation	91.00±04.65
w/o pseudo labeling method	92.13±05.90
About hyperparameter controlling strategy	P_{acc}
w/ fixed pseudo labeling	89.92±07.21
w/ linear dynamic pseudo labeling	92.46±04.95
w/ fixed γ for unsupervised pairwise loss in Eq. 17	89.47±10.22
PR-PL	**93.06±05.12**

V. D 我们进一步评估了采用不同参数设置时对模型性能的影响。本节的所有结果均基于 SEED 数据库，采用 EEG 信号情绪识别任务中常用的跨被试单次会话留一被试交叉验证评估协议。

A. 消融研究

我们进行消融研究，系统地探索所提出模型中不同组件的有效性，并检验其对整体性能的相应贡献。如表 X 所示，研究发现引入领域对抗训练可以显著提升目标域的情绪识别性能。当模型没有判别器且不包含目标域信息时，识别准确率从 93.06%降至 83.30%。这种显著下降表明个体差异问题对模型性能有重大影响，并突显了迁移学习在 aBCI 应用中的巨大潜力。 此外，结果表明，在源域和目标域上进行成对学习组合有利于模型性能，其中识别准确率提高了 6.35%（从 87.50%提高到 93.06%）。如果不考虑目标域上的特征可分性（无目标域成对学习），模型性能将从 93.06%下降到 88.81%。此外，在公式(5)中引入双线性变换矩阵 S 可以提高模型性能，其中识别准确率可以提高 4.75%（从 88.31%提高到 93.06%）。在与有无伪标签方法的比较中，将目标域的有效对逐步包含到训练过程中有助于学习稳定性，如性能从 92.13%提高到 93.06%所示。此外，还考察了不同伪标签方法的建模效果。当伪标签方法从固定变为线性动态时，相应的准确率从 89.92%提高到 92.46%。当采用基于非线性动态的自适应伪标签方法时，准确率进一步提高到 93.06%。 结果表明，基于非线性动态的自适应伪标签有助于筛选出有效的配对样本并提高模型的可训练性。对于公式(17)中给出的最终损失函数，我们提出在目标域上的无监督配对损失不使用固定的γ，而是随着训练轮次的增加逐渐更新γ，以防止模型在早期训练阶段学习失败，并平衡不同损失之间的关系。动态γ的优势也在消融研究中得到体现，其中识别准确率从 89.47%和 93.06%提升。使用所提出 PR-PL 中不同组件报告的结果存在显著差异，p 值小于 0.001。

B. 噪声标签的影响

为了进一步验证模型在噪声标签学习过程中的鲁棒性，我们随机将源标签污染η%的噪声，并在未知目标数据上测试相应的模型性能。具体来说，我们用随机生成的标签替换 Y 中η%的真实标签，并以监督学习的方式训练模型。然后，我们在目标域上测试训练好的模型性能。需要注意的是，噪声污染仅在源域进行，因为目标域需要用于模型评估。在实现中，η%的值分别调整为 10%、20%和 30%。相应的模型准确率及标准差为 89.22%±6.05%、88.39%±6.73%和 87.71%±5.02%。结果表明，随着标签噪声率从 10%增加到 30%，模型性能略有下降，下降率为 1.69%。这些结果证明了所提出的 PR-PL 是一种可靠的模型，具有更好的噪声标签容忍度。在未来的工作中，可以将最近提出的创新方法，如[62]和[63]，结合起来，以进一步消除 EEG 信号中的更普遍噪声，并提高模型在跨语料库应用中的稳定性。

C. 混淆矩阵

为了定性地研究模型在每个情绪类别中的性能，我们通过可视化分析混淆矩阵，并将结果与最新模型[8]、[24]、[50]进行比较。如图 2 所示，所有模型在区分积极情绪与其他情绪方面表现良好（识别率均超过 90%），但在区分消极情绪和中性情绪方面相对较差。例如，BiDANN[24]对中性情绪的识别率甚至低于 80%（76.72%）。与现有方法((a)、(b)和(c))相比，我们提出的模型能够提升模型的识别能力，尤其是在区分中性情绪和消极情绪方面。如图(d)所示，消极情绪、中性情绪和积极情绪的识别率分别为 92.10%、90.39%和 96.50%。其他对应于其他交叉验证评估协议（如被试内跨会话、被试内单会话、被试间跨会话留一被试外）的混淆矩阵结果在补充材料附录 I 中报告，可在网上获取。

D. 学习到的表示的可视化

为了从更直观的角度验证所提出模型的有效性，我们使用可视化方法呈现了源域和目标域的样本特征和交互特征

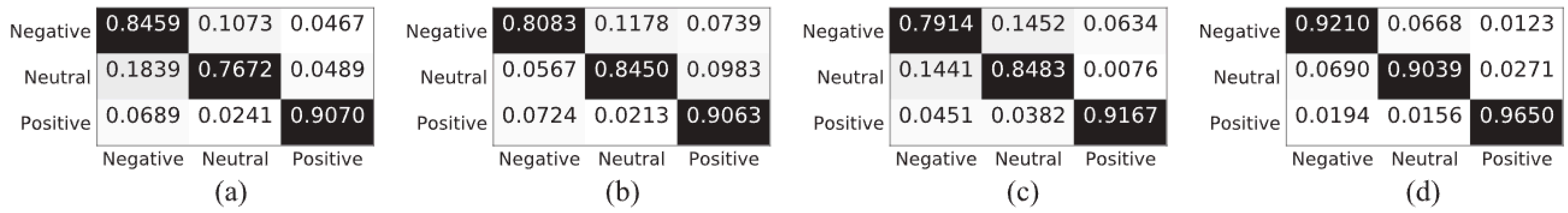

Fig. 2. Confusion matrices of different models. (a) BiDANN [24], (b) BiHDM [50], (c) RGNN [8], and (d) PR-PL.

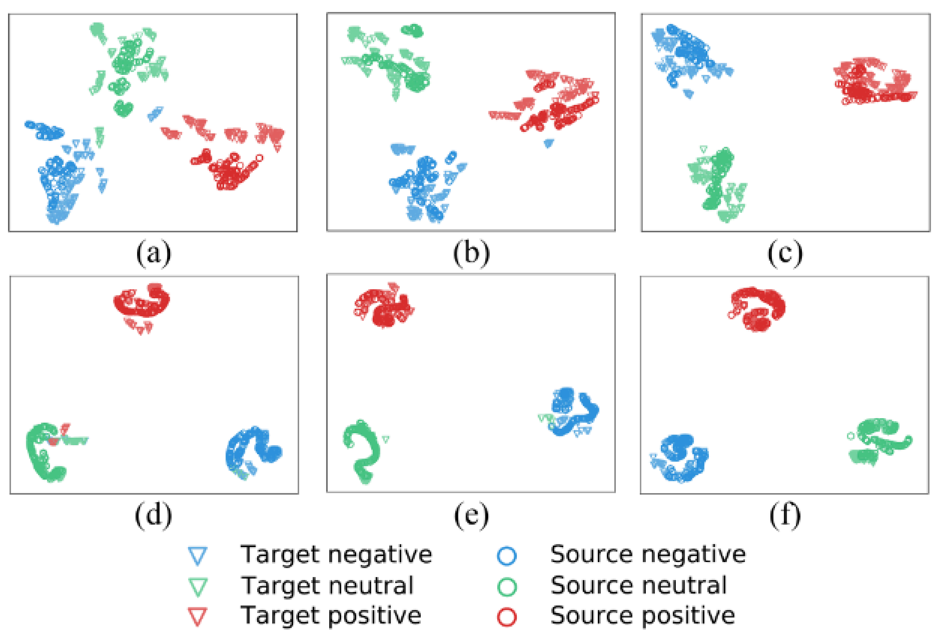

Fig. 3. T-SNE visualization of the learned features from the source and target domains under different model settings. (a), (b) and (c) are the sample features extracted by w/o pairwise learning on the source and target domains, w/o pairwise learning on the target domain, and PR-PL. (d), (e) and (f) are the interaction features extracted by w/o pairwise learning on the source and target domains, w/o pairwise learning on the target domain, and PR-PL.

T-SNE [64] in Fig. 3. Here, we randomly select 500 samples from the source domain and 500 samples from the target domain at one validation round for feature visualization. Compared to the representation learned by the other model settings (w/o pairwise learning on the source and target and w/o pairwise learning on the target), the representation learned by the proposed PR-PL forms more separated emotional clusters. Comparing the extracted sample features (c) and interaction features (f) by the proposed PR-PL, the separability of the extracted interaction features from different emotion classes is further enlarged and at the same time, the concentration of the feature distribution for each emotion is also improved. To better visualize the validity of the extracted features by the proposed PR-PL, a complete feature visualization in terms of the extracted features at the first hidden layer of the feature extractor $f(\cdot)$, sample features, prototype features, and interaction features at each validation round is reported in Appendix II of Supplementary Material, available online. Besides, to reveal the underlying emotion-related neural expression of the extracted features, we also conduct the topographic analysis on the obtained prototype features. Considering the paper length constraints, more implementation details with the corresponding results are presented in Appendix III of Supplementary Material, available online.

E. Computation Time and Incomplete Target Domain

To evaluate the time complexity of the proposed PR-PL, we estimate the computation time on all the target domain data. The average cost time of each data sample for emotion recognition is 0.00061 s, and the standard deviation is 0.00049 s. The transfer learning framework, on the other hand, aims to achieve knowledge transfer by adapting source knowledge to the unknown target domain. However, not all of the target domain data would be always accessible for model learning. Here, we further evaluate the robustness of the proposed PR-PL in the presence of an incomplete target domain. Specifically, we select the first K trials on the target domain as the validation set and the remaining trials $(N_{max} - K)$ as the test set. For the SEED database, N_{max} is equal to 15. To measure the model stability under different incomplete conditions, we adjust the K value from 3 to 12. To avoid biased class distribution, the step of K value is set to 3 (three-class classification task). In the implementation, only the source domain and the target domain's validation set are used for model learning. Then, the learned model is applied to the target domain's test set. In other words, the test set used for model validation is completely unseen during model training. The corresponding results are reported in Table XI, which shows that the proposed PR-PL could still be effective even in the case of incomplete target domain data.

TABLE XI
MODEL PERFORMANCE WITH INCOMPLETE TARGET DOMAIN

Validation Set (K trials)	Test Set (($N_{max} - K$) trials)	P_{acc}
3	12	85.07±05.84
6	9	85.35±07.72
9	6	86.88±07.90
12	3	88.58±08.98

VI. CONCLUSION

The article proposes a novel transfer learning framework with prototypical representation-based pairwise learning (PR-PL), that characterizes EEG data with prototypical representations and formulates the EEG-based emotion recognition tasks as pairwise learning. To estimate the model sensitive to the variants in subjects and sessions, we evaluate the proposed PR-PL on two well-known emotional databases (SEED and SEED-IV) under four different cross-validation evaluation protocols (within-subject cross-session, within-subject single-session, cross-subject cross-session, and cross-subject single-session) and compare it with the existing methods. Our extensive experimental results show that the proposed PR-PL achieves promising results on all four cross-validation evaluation protocols. Further, the model stability and generalizability under

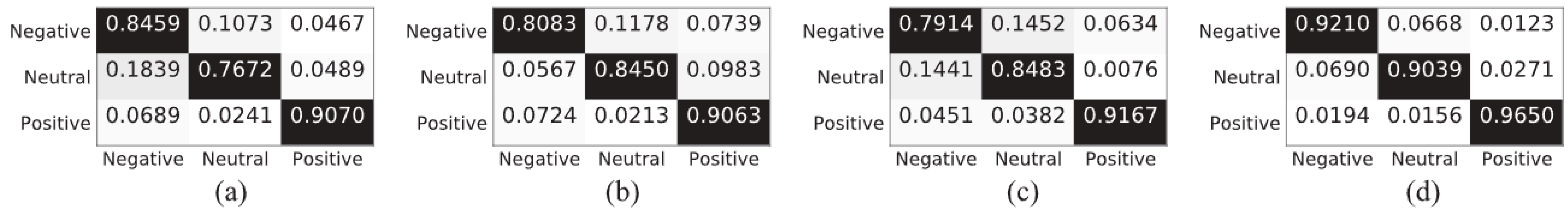

图 2. 不同模型的混淆矩阵。(a) BiDANN [24]，(b) BiHDM [50]，(c) RGNN [8]，以及 (d) PR-PL。

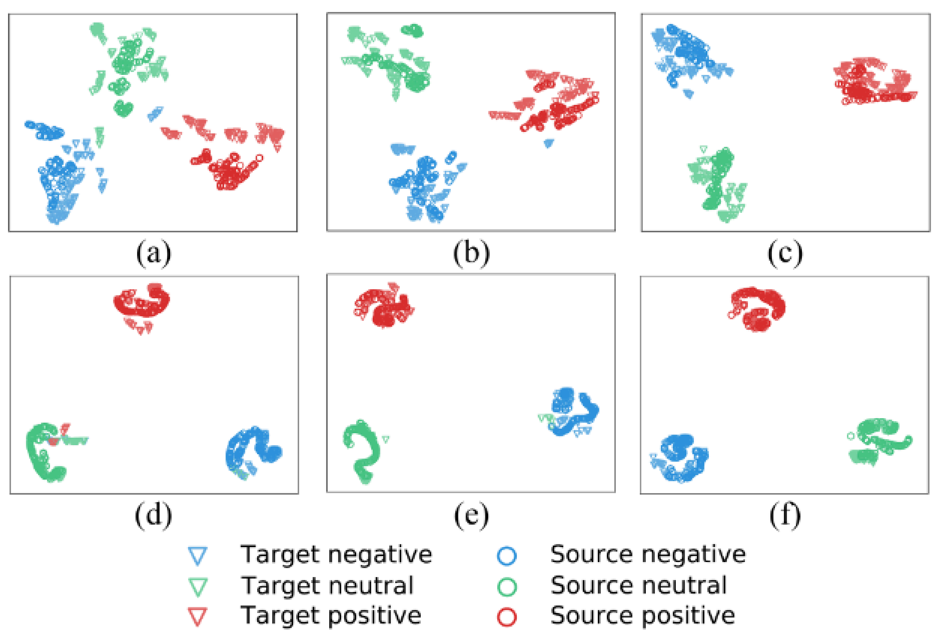

图 3. 在不同模型设置下，从源域和目标域学习到的特征的 T-SNE 可视化。(a)、(b) 和 (c) 是在源域和目标域上提取的无成对学习样本特征、在目标域上提取的无成对学习样本特征，以及 PR-PL。(d)、(e) 和 (f) 是在源域和目标域上提取的无成对学习交互特征、在目标域上提取的无成对学习交互特征，以及 PR-PL。

T-SNE [64] 在图 3 中。这里，我们在每一轮验证中随机选择 500 个源域样本和 500 个目标域样本进行特征可视化。与其他模型设置（源域和目标域不进行成对学习，目标域不进行成对学习）学习到的表示相比，PR-PL 学习到的表示形成了更分离的情感簇。通过比较 PR-PL 提取的样本特征（c）和交互特征（f），提取的交互特征在不同情感类别之间的可分性进一步扩大，同时每个情感的特征分布集中度也得到了改善。为了更好地可视化 PR-PL 提取特征的有效性，补充材料在线提供的附录 II 中报告了在特征提取器 f(·)的第一隐藏层提取的特征、样本特征、原型特征和每一轮验证中的交互特征的完整特征可视化。 此外，为了揭示提取特征背后的与情绪相关的神经表达，我们还对获得的原型特征进行了拓扑分析。考虑到论文篇幅限制，更多实现细节及相应结果在补充材料附录 III 中呈现，在线可获取。

E. 计算时间和非完整目标域

为了评估 PR-PL 的时间复杂度，我们在所有目标域上估计了计算时间

表 XI

MPW I TD

Validation Set (K trials)	Test Set (($N_{max} - K$) trials)	P_{acc}
3	12	85.07±05.84
6	9	85.35±07.72
9	6	86.88±07.90
12	3	88.58±08.98

数据。情绪识别每个数据样本的平均成本时间为 0.00061 秒，标准差为 0.00049 秒。迁移学习框架旨在通过将源知识适应未知目标域来实现知识迁移。然而，并非所有目标域数据都始终可用于模型学习。在此，我们进一步评估了在目标域不完整的情况下所提出的 PR-PL 的鲁棒性。具体而言，我们选择目标域上的前 K 次试验作为验证集，其余的试验（N-K）作为测试集。对于 SEED 数据库，N 等于 15。为了测量模型在不同不完整条件下的稳定性，我们将 K 值从 3 调整到 12。为了避免类别分布的偏差，K 值的步长设置为 3（三分类任务）。在实现中，仅使用源域和目标域的验证集进行模型学习。然后，将学习到的模型应用于目标域的测试集。换句话说，用于模型验证的测试集在模型训练期间完全未出现过。 相应的结果报告在表 XI 中，该表显示即使在不完整的源域数据情况下，所提出的 PR-PL 仍然有效。

VI. C 该文章提出了一种基于原型表示的成对学习（PRPL）的新型迁移学习框架，该框架使用原型表示来表征脑电图（EEG）数据，并将基于 EEG 的情绪识别任务表述为成对学习。为了评估模型对受试者和会话中变化的敏感性，我们在四个不同的交叉验证评估协议（受试者内跨会话、受试者内单会话、受试者间跨会话和受试者间单会话）下，在两个著名的情绪数据库（SEED 和 SEED-IV）上评估所提出的 PR-PL，并将其与现有方法进行比较。我们的广泛实验结果表明，所提出的 PR-PL 在所有四个交叉验证评估协议上均取得了令人满意的结果。此外，该模型在

various experimental settings are fully validated. These experimental findings indicate the superiority of the proposed PR-PL in tackling individual differences and noisy labeling issues in aBCI systems.

References

[1] S. Siddharth, T.-P. Jung, and T. J. Sejnowski, "Utilizing deep learning towards multi-modal bio-sensing and vision-based affective computing," *IEEE Trans. Affect. Comput.*, vol. 13, no. 1, pp. 96–107, First Quarter 2022.

[2] X. Hu, J. Chen, F. Wang, and D. Zhang, "Ten challenges for EEG-based affective computing," *Brain Sci. Adv.*, vol. 5, no. 1, pp. 1–20, 2019.

[3] W. Hu, G. Huang, L. Li, L. Zhang, Z. Zhang, and Z. Liang, "Video-triggered EEG-emotion public databases and current methods: A survey," *Brain Sci. Adv.*, vol. 6, no. 3, pp. 255–287, 2020.

[4] V. Jayaram, M. Alamgir, Y. Altun, B. Scholkopf, and M. Grosse-Wentrup, "Transfer learning in brain-computer interfaces," *IEEE Comput. Intell. Mag.*, vol. 11, no. 1, pp. 20–31, Feb. 2016.

[5] J. Li, S. Qiu, Y.-Y. Shen, C.-L. Liu, and H. He, "Multisource transfer learning for cross-subject EEG emotion recognition," *IEEE Trans. Cybern.*, vol. 50, no. 7, pp. 3281–3293, Jul. 2020.

[6] Y. Li, W. Zheng, L. Wang, Y. Zong, and Z. Cui, "From regional to global brain: A novel hierarchical spatial-temporal neural network model for EEG emotion recognition," *IEEE Trans. Affect. Comput.*, vol. 13, no. 2, pp. 568–578, Second Quarter 2022.

[7] H. Cui, A. Liu, X. Zhang, X. Chen, K. Wang, and X. Chen, "EEG-based emotion recognition using an end-to-end regional-asymmetric convolutional neural network," *Knowl.-Based Syst.*, vol. 205, 2020, Art. no. 106243.

[8] P. Zhong, D. Wang, and C. Miao, "EEG-based emotion recognition using regularized graph neural networks," *IEEE Trans. Affect. Comput.*, vol. 13, no. 3, pp. 1290–1301, Third Quarter 2022.

[9] X. Gu et al., "EEG-based brain-computer interfaces (BCIs): A survey of recent studies on signal sensing technologies and computational intelligence approaches and their applications," *IEEE/ACM Trans. Comput. Biol. Bioinf.*, vol. 18, no. 5, pp. 1645–1666, Sep./Oct. 2021.

[10] O. Özdenizci, Y. Wang, T. Koike-Akino, and D. Erdoğmuş, "Adversarial deep learning in EEG biometrics," *IEEE Signal Process. Lett.*, vol. 26, no. 5, pp. 710–714, May 2019.

[11] O. Özdenizci, Y. Wang, T. Koike-Akino, and D. Erdoğmuş, "Learning invariant representations from EEG via adversarial inference," *IEEE Access*, vol. 8, pp. 27074–27085, 2020.

[12] D. Bethge et al., "Domain-invariant representation learning from EEG with private encoders," 2022, *arXiv:2201.11613*.

[13] W.-L. Zheng and B.-L. Lu, "Investigating critical frequency bands and channels for EEG-based emotion recognition with deep neural networks," *IEEE Trans. Auton. Ment. Develop.*, vol. 7, no. 3, pp. 162–175, Sep. 2015.

[14] Y. Jia, M. Salzmann, and T. Darrell, "Factorized latent spaces with structured sparsity," in *Proc. Adv. Neural Inf. Process. Syst.*, 2010, pp. 982–990.

[15] H. Bao, G. Niu, and M. Sugiyama, "Classification from pairwise similarity and unlabeled data," in *Proc. Int. Conf. Mach. Learn.*, PMLR, 2018, pp. 452–461.

[16] H. Bao, T. Shimada, L. Xu, I. Sato, and M. Sugiyama, "Similarity-based classification: Connecting similarity learning to binary classification," 2020, *arXiv:2006.06207*.

[17] C.-C. Hsu, Y.-X. Zhuang, and C.-Y. Lee, "Deep fake image detection based on pairwise learning," *Appl. Sci.*, vol. 10, no. 1, pp. 1–14, 2020.

[18] P. Zhuang, Y. Wang, and Y. Qiao, "Learning attentive pairwise interaction for fine-grained classification," in *Proc. AAAI Conf. Artif. Intell.*, 2020, pp. 13130–13137.

[19] L. Yao, S. Li, Y. Li, M. Huai, J. Gao, and A. Zhang, "Representation learning for treatment effect estimation from observational data," in *Proc. Adv. Neural Inf. Process. Syst.*, 2018, pp. 2638–2648.

[20] S. J. Pan, I. W. Tsang, J. T. Kwok, and Q. Yang, "Domain adaptation via transfer component analysis," *IEEE Trans. Neural Netw.*, vol. 22, no. 2, pp. 199–210, Feb. 2011.

[21] W.-L. Zheng and B.-L. Lu, "Personalizing EEG-Based affective models with transfer learning," in *Proc. 25th Int. Joint Conf. Artif. Intell.*, AAAI Press, 2016, pp. 2732–2738.

[22] Y. Ganin et al., "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2096–2030, 2016.

[23] Y.-M. Jin, Y.-D. Luo, W.-L. Zheng, and B.-L. Lu, "EEG-based emotion recognition using domain adaptation network," in *Proc. Int. Conf. Orange Technol.*, 2017, pp. 222–225.

[24] Y. Li, W. Zheng, Y. Zong, Z. Cui, T. Zhang, and X. Zhou, "A bi-hemisphere domain adversarial neural network model for EEG emotion recognition," *IEEE Trans. Affect. Comput.*, vol. 12, no. 2, pp. 494–504, Second Quarter 2021.

[25] X. Du et al., "An efficient LSTM network for emotion recognition from multichannel EEG signals," *IEEE Trans. Affect. Comput.*, vol. 13, no. 3, pp. 1528–1540, Third Quarter 2022.

[26] Y. Luo, S. Y. Zhang, W. L. Zheng, and B. L. Lu, "WGAN domain adaptation for EEG-based emotion recognition," in *Proc. Int. Conf. Neural Inf. Process.*, 2018, pp. 275–286.

[27] J. Li, S. Qiu, C. Du, Y. Wang, and H. He, "Domain adaptation for EEG emotion recognition based on latent representation similarity," *IEEE Trans. Cogn. Devel. Syst.*, vol. 12, no. 2, pp. 344–353, Jun. 2020.

[28] Y. Wang et al., "EEG-based emotion recognition with prototype-based data representation," in *Proc. IEEE 41st Annu. Int. Conf. Eng. Med. Biol. Soc.*, 2019, pp. 684–689.

[29] Y. Wang, S. Qiu, X. Ma, and H. He, "A prototype-based SPD matrix network for domain adaptation EEG emotion recognition," *Pattern Recognit.*, vol. 110, 2021, Art. no. 107626.

[30] F. Shen, Y. Peng, G. Dai, B. Lu, and W. Kong, "Coupled projection transfer metric learning for cross-session emotion recognition from EEG," *Systems*, vol. 10, no. 2, pp. 1–19, 2022.

[31] M. Salzmann, C. H. Ek, R. Urtasun, and T. Darrell, "Factorized orthogonal latent spaces," *J. Mach. Learn. Res.*, vol. 9, pp. 701–708, 2010.

[32] K. Bousmalis, G. Trigeorgis, N. Silberman, D. Krishnan, and D. Erhan, "Domain separation networks," in *Proc. 30th Int. Conf. Neural Inf. Process. Syst.*, 2016, pp. 343–351.

[33] P. O. Pinheiro, "Unsupervised domain adaptation with similarity learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 8004–8013.

[34] J. Chang, L. Wang, G. Meng, S. Xiang, and C. Pan, "Deep adaptive image clustering," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2017, pp. 5880–5888.

[35] W.-L. Zheng, W. Liu, Y. Lu, B.-L. Lu, and A. Cichocki, "EmotionMeter: A multimodal framework for recognizing human emotions," *IEEE Trans. Cybern.*, vol. 49, no. 3, pp. 1110–1122, Mar. 2019.

[36] L.-C. Shi and B.-L. Lu, "Off-line and on-line vigilance estimation based on linear dynamical system and manifold learning," in *Proc. Annu. Int. Conf. IEEE Eng. Med. Biol.*, 2010, pp. 6587–6590.

[37] D. Wu, Y. Xu, and B.-L. Lu, "Transfer learning for EEG-based brain–computer interfaces: A review of progress made since 2016," *IEEE Trans. Cogn. Devel. Syst.*, vol. 14, no. 1, pp. 4–19, Mar. 2022.

[38] W. Li, W. Huan, B. Hou, Y. Tian, Z. Zhang, and A. Song, "Can emotion be transferred?–A review on transfer learning for EEG-based emotion recognition," *IEEE Trans. Cogn. Devel. Syst.*, vol. 14, no. 3, pp. 833–846, Sep. 2022.

[39] W.-L. Zheng, Y.-Q. Zhang, J.-Y. Zhu, and B.-L. Lu, "Transfer components between subjects for EEG-based emotion recognition," in *Proc. Int. Conf. Affect. Comput. Intell. Interaction*, 2015, pp. 917–922.

[40] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

[41] D. Coomans and D. L. Massart, "Alternative k-nearest neighbour rules in supervised pattern recognition: Part 1. k-nearest neighbour classification by using alternative voting rules," *Analytica Chimica Acta*, vol. 136, pp. 15–27, 1982. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0003267001953590

[42] B. Sun, J. Feng, and K. Saenko, "Return of frustratingly easy domain adaptation," in *Proc. 30th AAAI Conf. Artif. Intell.*, AAAI Press, 2016, pp. 2058–2065.

[43] B. Fernando, A. Habrard, M. Sebban, and T. Tuytelaars, "Unsupervised visual domain adaptation using subspace alignment," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2013, pp. 2960–2967.

[44] B. Gong, Y. Shi, F. Sha, and K. Grauman, "Geodesic flow kernel for unsupervised domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2012, pp. 2066–2073.

[45] M. Long, Y. Cao, J. Wang, and M. Jordan, "Learning transferable features with deep adaptation networks," in *Proc. Int. Conf. Mach. Learn.*, PMLR, 2015, pp. 97–105.

[46] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, and T. Darrell, "Deep domain confusion: Maximizing for domain invariance," 2014, *arXiv:1412.3474*. [Online]. Available: http://arxiv.org/abs/1412.3474

各种实验设置均得到充分验证。这些实验结果表明，所提出的 PR-PL 在处理 aBCI 系统中的个体差异和噪声标签问题方面具有优越性。

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[1] S. Siddharth, T.-P. Jung, 和 T. J. Sejnowski, “利用深度学习进行多模态生物传感和基于视觉的情感计算,” IEEE 情感计算汇刊, 第 13 卷, 第 1 期, 第 96-107 页, 2022 年第一季度.

[2] X. Hu, J. Chen, F. Wang, 和 D. Zhang, “基于脑电图的情感计算的十个挑战,” 大脑科学进展, 第 5 卷, 第 1 期, 第 1-20 页, 2019 年. [3] W. Hu, G. Huang, L. Li, L. Zhang, Z. Zhang, 和 Z. Liang, “视频触发的脑电图-情感公共数据库和当前方法: 综述,” Brain Sci. Adv., vol. 6, no. 3, pp. 255–287, 2020.

[4] V. Jayaram, M. Alamgir, Y. Altun, B. Scholkopf, 和 M. Grosse-Wentrup, “脑机接口中的迁移学习,” IEEE 计算机智能 第 11 卷，第 1 期，第 20-31 页，2016 年 2 月。

[5] J. Li, S. Qiu, Y.-Y. Shen, C.-L. Liu, 和 H. He, “多源迁移学习用于跨被试 EEG 情绪识别,” IEEE 控制论汇刊, 第 50 卷, 第 7 期, pp. 3281–3293, 2020 年 7 月. [6] Y. Li, W. Zheng, L. Wang, Y. Zong, 和 Z. Cui, “从区域到全局大脑：一种用于 EEG 情绪识别的新型层次时空神经网络模型,” IEEE 情感计算汇刊, 第 13 卷, 第 2 期, pp. 568– 578, 2022 年第二季度. [7] H. Cui, A. Liu, X. Zhang, X. Chen, K. Wang, 和 X. Chen, “基于 EEG 的情绪识别，使用端到端的区域非对称卷积神经网络,” 知识系统, 第 205 期, 2020 年, 文章编号 106243. [8] P. Zhong, D. Wang, 和 C. Miao, “基于 EEG 的情绪识别，使用正则化图神经网络,” IEEE 情感计算汇刊, 第 13 卷, 第 3 期, pp. 1290–1301, 2022 年第三季度. [9] X. Gu 等, “基于 EEG 的脑机接口（BCIs）：信号传感技术和计算智能方法及其应用的最新研究综述,” IEEE/ACM 计算汇刊.

生物生物学与生物信息学，第 18 卷，第 5 期，第 1645-1666 页，2021 年 9 月/10 月。

[10] O. Özdenizci, Y. Wang, T. Koike-Akino, 和 D. Erdo˘gmu¸s, “对抗深度学习在脑电图生物识别中的应用，”IEEE 信号处理信札，第 26 卷，第 5 期，第 710-714 页，2019 年 5 月。

[11] O. Özdenizci, Y. Wang, T. Koike-Akino, and D. Erdo˘gmu¸s, “学习通过对抗推理从脑电图信号中获取变体表示，”IEEE Access，卷. 8，页. 27074–27085，2020。

[12] D. Bethge 等人，“基于私有编码器从脑电图学习领域不变表示”，2022 年，arXiv:2201.11613。 [13] W.-L. Zheng 和 B.-L. Lu，“研究用于基于脑电图的深度神经网络情绪识别的关键频段和通道”，

IEEE Transactions on Autonomous Mental Development, vol. 7, no. 3, pp. 162–175, Sep. 2015.

[14] Y. Jia, M. Salzmann, and T. Darrell, “因子分解的潜在空间与结构化稀疏性,” 在《高级神经信息处理系统会议录》2010 年, 第 982-990 页。 [15] H. Bao, G. Niu, 和 M. Sugiyama, “基于成对相似性的分类” 以及无标签数据，”在 2018 年机器学习国际会议论文集中，PMLR 出版社，第 452-461 页。

[16] Bao H., Shimada T., Xu L., Sato I. 和 Sugiyama M.,“基于相似性的分类: 连接相似性学习与二元分类,” 2020, arXiv:2006.06207.

[17] C.-C. Hsu, Y.-X. Zhuang, 和 C.-Y. Lee, “基于成对学习的深度伪造图像检测,” 应用科学, 第 10 卷, 第 1 期, 第 1-14 页, 2020. [18] P. Zhuang, Y. Wang, 和 Y. Qiao, “学习注意成对交互” 用于细粒度分类，”在 2020 年 AAAI 人工智能会议论文集中，第 13130-13137 页。

[19] L. 姚 S. 李 Y. 李 M. 汪 J. 高 A. 张，“从观察数据中学习治疗效果表示，”在会议论文集中 《高级神经信息处理系统》，2018 年，第 2638-2648 页。

[20] S. J. Pan, I. W. Tsang, J. T. Kwok 和 Q. Yang，“领域自适应通过迁移成分分析，”IEEE 神经网络汇刊，第 22 卷，第 2 期，第 199-210 页，2011 年 2 月。

[21] 郑伟立和陆北立，“个性化基于脑电的情感模型 “基于迁移学习的原型表示对偶学习框架——用于脑电图信号的情绪识别”，第 25 届国际人工智能联合会议论文集，AAAI 出版社，2016 年，第 2732-2738 页。

[22] Y. Ganin 等人，“神经网络的领域对抗训练”，机器学习杂志 Learn. Res.，卷 17, 第 1 期, 第 2096–2030 页, 2016.

[23] 金亚敏, 罗亚东, 郑伟立, 和 陆伯伦,“基于脑电的情绪 使用领域自适应网络的情感识别，”在 2017 年国际橙技术会议上，第 222-225 页。

[24] 李岩，郑伟，宗杨，崔哲，张涛，周翔，“一种用于脑电信号情绪识别的双半球域对抗神经网络模型，” IEEE Trans. Affect. Comput., vol. 12, no. 2, pp. 494–504, 2021 年第二季度。

[25] 杜雪等，“一种用于情绪识别的高效 LSTM 网络” 多通道脑电图信号，”IEEE 情感计算汇刊，第 13 卷，第 3 期，第 1528-1540 页，2022 年第三季度。

[26] 罗宇，张思宇，郑伟立，和陆冰蓝，“WGAN 域 基于脑电图的情绪识别的适应性，”在 2018 年国际神经信息处理会议论文集中，第 275-286 页。

[27] 李杰，邱松，杜超，王宇，和何海，“领域自适应 基于潜在表示相似性的脑电图情绪识别，”IEEE 认知发展系统汇刊，第 12 卷，第 2 期，第 344-353 页，2020 年 6 月。

[28] 王勇等，“基于脑电信号的基于原型数据表示的情绪识别，”在 IEEE 第 41 届年度国际工程医学与生物学会议论文集中 Soc., 2019, pp. 684–689.

[29] 王勇，邱珊，马晓，和何海，“基于原型的 SPD 矩阵网络” 用于领域自适应的 EEG 情绪识别的工作，”模式识别，vol. 110，2021 年，第 107626 号文章。

[30] F. Shen, Y. Peng, G. Dai, B. Lu, 和 W. Kong, “耦合投影迁移 从脑电图信号进行跨会话情绪识别的度量学习，” Systems，第 10 卷，第 2 期，第 1-19 页，2022 年。

[31] M. Salzmann, C. H. Ek, R. Urtasun, 和 T. Darrell, “因子分解正交潜在空间,” 机器学习研究杂志, 卷 9, 页 701–708, 2010. [32] K. Bousmalis, G. Trigeorgis, N. Silberman, D. Krishnan, 和 D. Erhan, “领域分离网络,” 在第 30 届国际神经信息处理会议论文集中 Syst., 2016, pp. 343–351.

[33] P. O. Pinheiro, “无监督域适应与相似性学习” “ing, ”在 IEEE/CVF 计算机视觉与模式识别会议论文集中，2018 年，第 8004-8013 页。

[34] J. Chang, L. Wang, G. Meng, S. Xiang, 和 C. Pan, “深度自适应图像聚类,” 在 IEEE 国际计算机视觉会议论文集，2017 年, pp. 5880–5888. [35] W.-L. Zheng, W. Liu, Y. Lu, B.-L. Lu, 和 A. Cichocki, “EmotionMeter: 一种用于识别人类情绪的多模态框架,” IEEE 交易. 网络，第 49 卷，第 3 期，第 1110-1122 页，2019 年 3 月。

[36] L.-C. Shi 和 B.-L. Lu, “基于线性动力系统和流形学习的离线与在线警觉度估计,” 在《Proc. Annu. Int.》 IEEE 工程医学与生物学学会会议录，2010，第 6587–6590 页。

[37] D. Wu, Y. Xu, 和 B.-L. Lu, “基于脑机接口的脑电信号迁移学习：2016 年以来的进展综述,” IEEE Trans. 认知发展系统，第 14 卷，第 1 期，第 4-19 页，2022 年 3 月。

[38] 李伟，黄伟，侯斌，田宇，张哲，宋安，“情绪 如何迁移？—基于脑电图的情绪识别迁移学习综述，”IEEE 认知发展系统汇刊，第 14 卷，第 3 期，第 833-846 页，2022 年 9 月。

[39] 郑伟立，张一齐，朱建宇，和 陆北林，“基于脑电信号的情绪识别中跨被试转移组件，” 在 国际会议论文集 情感计算与人工智能交互，2015，第 917-922 页。

[40] L. Breiman，“随机森林，” 机器学习, 卷. 45, 第. 1, 页. 5–32, 2001.

[41] D. Coomans 和 D. L. Massart，“替代 k 近邻规则在 监督模式识别：第一部分。使用替代投票规则的 k 近邻分类，”分析化学学报，卷. 136，页. 15–27，1982。[在线]. 可获得: https://www.sciencedirect.com/ science/article/pii/S0003267001953590

[42] B. Sun, J. Feng, and K. Saenko, “令人沮丧的简单领域 “适应”，在《第 30 届 AAAI 人工智能会议论文集》，AAAI 出版社，2016 年，第 2058-2065 页。

[43] B. Fernando, A. Habrard, M. Sebban 和 T. Tuytelaars, “基于子空间对齐的无监督视觉领域自适应,” 在 IEEE 国际会议论文集 《计算机视觉会议论文集》，2013 年，第 2960-2967 页。

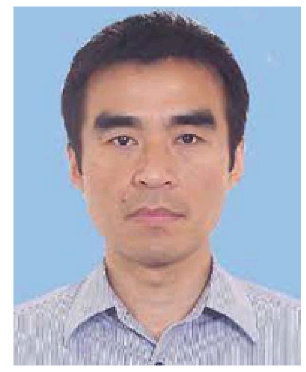
[44] B. Gong, Y. Shi, F. Sha, and K. Grauman, “Geodesic flow kernel for un- "监督域适应," 在 IEEE 计算机视觉与模式识别会议论文集中, 2012, pp. 2066–2073.

[45] M. Long, Y. Cao, J. Wang, 和 M. Jordan, “学习可迁移特征 “基于深度自适应网络，”在 2015 年机器学习国际会议论文集中，PMLR 出版社，第 97-105 页。

[46] E. Tzeng, J. Hoffman, N. Zhang, K. Saenko, 和 T. Darrell, “深度域混淆: 最大化域不变性,” 2014, arXiv:1412.3474. [在线]. 可获得: http://arxiv.org/abs/1412.3474

[47] B. Sun and K. Saenko, “Deep CORAL: Correlation alignment for deep domain adaptation,” in *Proc. Eur. Conf. Comput. Vis.*, G. Hua and H. Jégou, Eds., Cham: Springer International Publishing, 2016, pp. 443–450.
[48] H. Chen, Z. Li, M. Jin, and J. Li, “MEERNet: Multi-source EEG-based emotion recognition network for generalization across subjects and sessions,” in *Proc. IEEE 43rd Annu. Int. Conf. Eng. Med. Biol. Soc.*, 2021, pp. 6094–6097.
[49] T. Song, W. Zheng, P. Song, and Z. Cui, “EEG emotion recognition using dynamical graph convolutional neural networks,” *IEEE Trans. Affect. Comput.*, vol. 11, no. 3, pp. 532–541, Third Quarter 2020.
[50] Y. Li et al., “A novel bi-hemispheric discrepancy model for EEG emotion recognition,” *IEEE Trans. Cogn. Devel. Syst.*, vol. 13, no. 2, pp. 354–367, Jun. 2021.
[51] Y. Li, W. Zheng, Z. Cui, Y. Zong, and S. Ge, “EEG emotion recognition based on graph regularized sparse linear regression,” *Neural Process. Lett.*, vol. 49, pp. 1–17, Apr. 2019.
[52] W. Zheng, “Multichannel EEG-based emotion recognition via group sparse canonical correlation analysis,” *IEEE Trans. Cogn. Devel. Syst.*, vol. 9, no. 3, pp. 281–290, Sep. 2017.
[53] T. Zhang, W. Zheng, Z. Cui, Y. Zong, and Y. Li, “Spatial–Temporal recurrent neural network for emotion recognition,” *IEEE Trans. Cybern.*, vol. 49, no. 3, pp. 839–847, Mar. 2019.
[54] J. Suykens and J. Vandewalle, “Least squares support vector machine classifiers,” *Neural Process. Lett.*, vol. 9, no. 3, pp. 293–300, 1999.
[55] J. Zhu, A. Arbor, and T. Hastie, “Multi-class AdaBoost,” *Statist. Interface*, vol. 2, no. 3, pp. 349–360, 2006.
[56] H. Li, Y.-M. Jin, W.-L. Zheng, and B.-L. Lu, “Cross-subject emotion recognition using deep adaptation networks,” in *Proc. Int. Conf. Neural Inf. Process.*, L. Cheng, A. C. S. Leung, and S. Ozawa, Eds., Cham: Springer International Publishing, 2018, pp. 403–413.
[57] B.-Q. Ma, H. Li, W.-L. Zheng, and B.-L. Lu, “Reducing the subject variability of EEG signals with adversarial domain generalization,” in *Proc. Int. Conf. Neural Inf. Process.*, T. Gedeon, K. W. Wong, and M. Lee, Eds., Cham: Springer International Publishing, 2019, pp. 30–42.
[58] Y. Peng, W. Wang, W. Kong, F. Nie, B.-L. Lu, and A. Cichocki, “Joint feature adaptation and graph adaptive label propagation for cross-subject emotion recognition from EEG signals,” *IEEE Trans. Affect. Comput.*, vol. 13, no. 4, pp. 1941–1958, Fourth Quarter 2022.
[59] Z. Li et al., “Dynamic domain adaptation for class-aware cross-subject and cross-session EEG emotion recognition,” *IEEE J. Biomed. Health Informat.*, vol. 26, no. 12, pp. 5964–5973, Dec. 2022.
[60] H. Chen, M. Jin, Z. Li, C. Fan, J. Li, and H. He, “MS-MDA: Multisource marginal distribution adaptation for cross-subject and cross-session EEG emotion recognition,” *Front. Neurosci.*, vol. 15, pp. 1–14, 2021.
[61] L. Zhu et al., “Multisource wasserstein adaptation coding network for EEG emotion recognition,” *Biomed. Signal Process. Control*, vol. 76, 2022, Art. no. 103687. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S1746809422002099
[62] X. Xiao, M. Xu, J. Jin, Y. Wang, T.-P. Jung, and D. Ming, “Discriminative canonical pattern matching for single-trial classification of ERP components,” *IEEE Trans. Biomed. Eng.*, vol. 67, no. 8, pp. 2266–2275, Aug. 2020.
[63] J. Jin, Z. Wang, R. Xu, C. Liu, X. Wang, and A. Cichocki, “Robust similarity measurement based on a novel time filter for SSVEPs detection,” *IEEE Trans. Neural Netw. Learn. Syst.*, to be published, doi: 10.1109/TNNLS.2021.3118468.
[64] V. D. M. Laurens and G. Hinton, “Visualizing data using t-SNE,” *J. Mach. Learn. Res.*, vol. 9, no. 2605, pp. 2579–2605, 2008.

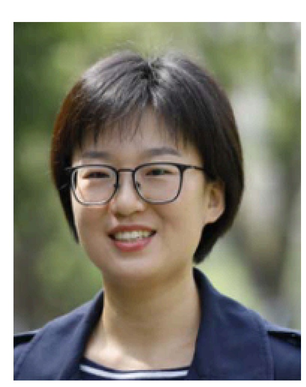
Rushuang Zhou is currently working toward the PhD degree with the Department of Biomedical Engineering, City University of Hong Kong, Hong Kong. His research interests include affective computing and transfer learning.



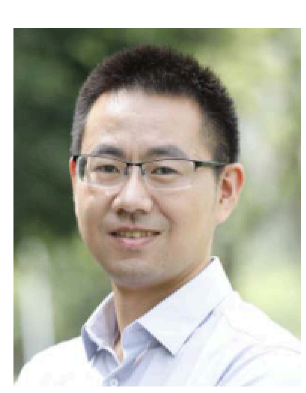
Zhiguo Zhang (Member, IEEE) received the BEng degree from Tianjin University, in 2000, the MPhil degree from the University of Science and Technology of China, in 2003, and the PhD degree from the University of Hong Kong, in 2008. He is now a professor with the Institute of Computing and Intelligence, Harbin Institute of Technology, Shenzhen, China. His research interests include biomedical signal processing, neural engineering, and brain-inspired computation.

Hong Fu received the PhD degree from The Hong Kong Polytechnic University, in 2007. She is now an assistant professor with the Department of Mathematics and Information Technology, The Education University of Hong Kong, Hong Kong. Her research interests include computer vision algorithms and artificial intelligence in healthcare and education.

Li Zhang received the PhD degree from the Department of Electrical and Electronics, University of Hong Kong, in 2017 and joined the School of Biomedical Engineering, Medical School, Shenzhen University, in 2018 as an associate researcher. His current research interests mainly focus on biomedical signal processing, numerical optimization, and imaging genetics.



Linling Li received the PhD degree from the University of Chinese Academy of Sciences, China, in 2017, and then received the postdoctoral training with Shenzhen University, China. She is now an associate researcher with the School of Biomedical Engineering, Medical School, Shenzhen University, China. Her current research interests focus on the development of EEG neurofeedback technique and related neuroimaging mechanisms.



Gan Huang received the PhD degree from Shanghai Jiao Tong University, China, in 2013, and then received his postdoctoral training with Université Catholique de Louvain in Belgium. He is now an assistant professor with the School of Biomedical Engineering, Shenzhen University, China. He has published several high-quality papers on journals including *Neuroimage*, *Neural Network*, and *Neurocomputing*. His current research interests focus on neural modulation, brain-computer interface and neuroprosthesis.

[47] B. Sun 和 K. Saenko, “Deep CORAL: 深度域适应的相关性对齐,” 在 欧洲计算机视觉会议论文集, G. Hua 和 H.
Jégou, Eds., 查姆: 斯普林格国际出版公司, 2016, 第 443-450 页。
[48] H. Chen, Z. Li, M. Jin, 和 J. Li, “MEERNet: 基于多源脑电信号的多源脑电信号模型”用于跨被试和会话泛化的情绪识别网络，”IEEE 第 43 届年度国际工程医学与生物学学会会议论文集，2021 年，第 6094-6097 页。

[49] T.宋, W.郑, P.宋, 和 Z.崔, “使用动态图卷积神经网络的脑电图情绪识别,” IEEE Affect. Trans.
计算，第 11 卷，第 3 期，第 532-541 页，2020 年第三季度。
[50] 李勇等，“一种新型双半球差异模型用于脑电图情绪识别”
识别，” IEEE 交易. 认知发展系统，卷. 13，第. 2，第. 354–367，2021 年 6 月。
[51] 李岩，郑伟，崔志，宗岩，和葛松，“脑电图情绪识别
基于图正则化稀疏线性回归，”神经过程快报，vol. 49, pp. 1–17, 2019 年 4 月。
[52] W. 郑敏，“基于多通道脑电图的情绪识别”
稀疏典型相关分析，”IEEE 认知发展系统汇刊，vol. 9, no. 3, pp. 281–290, Sep. 2017.
[53] 张涛，郑伟，崔峥，宗宇，李岩，“时空
用于情绪识别的循环神经网络，”IEEE 资讯科学汇刊，第 49 卷，第 3 期，第 839-847 页，2019 年 3 月。
[54] J. Suykens 和 J. Vandewalle, “最小二乘支持向量机分类器,” 神经处理快报, 卷. 9, 第. 3, pp. 293–300, 1999. [55] J. Zhu, A. Arbor, 和 T. Hastie, “多类 AdaBoost,” 统计界面, 卷. 2, 第. 3, pp. 349–360, 2006. [56] H. Li, Y.-M. Jin, W.-L. Zheng, 和 B.-L. Lu, “跨被试情绪识别使用深度适应网络,” 在 国际神经信息会议论文集.

Process., L. Cheng, A. C. S. Leung, 和 S. Ozawa, 编, Cham: Springer International Publishing, 2018, 第 403–413 页。
[57] B.-Q. Ma, H. Li, W.-L. Zheng, 和 B.-L. Lu, “减少被试
EEG 信号在对抗领域泛化中的变异性，”在 T. Gedeon、K. W. Wong 和 M. 的《国际神经信息处理会议论文集》中
李，编， Cham: Springer International Publishing, 2019, pp. 30–42.
[58] Y. Peng, W. Wang, W. Kong, F. Nie, B.-L. Lu, 和 A. Cichocki, “联合
特征适应和图自适应标签传播用于从脑电图信号进行跨被试情绪识别,” IEEE 情感计算汇刊, 第 13 卷, 第 4 期, pp. 1941–1958, 2022 年第四季度。

[59] Z. Li 等人, “动态域适应用于类感知跨被试
以及跨会话脑电图情绪识别，”IEEE 生物医学与健康信息杂志，第 26 卷，第 12 期，第 5964-5973 页，2022 年 12 月。
[60] H. Chen, M. Jin, Z. Li, C. Fan, J. Li, 和 H. He, “MS-MDA: Multisource
跨被试和跨会话的 EEG 情绪识别的边缘分布自适应,” Front. Neurosci., vol. 15, pp. 1–14, 2021.
[61] L. Zhu 等人，“多源 Wasserstein 自适应编码网络”
脑电图情绪识别，”生物医学信号处理与控制，卷 76，2022 年，文章编号 103687。[在线]。可获取: https://www.sciencedirect.com/science/article/pii/S1746809422002099
[62] X. 小晓, M. 许, J. 金, Y. 王晓, T.-P. 容, 和 D. 明, “区分-
基于脑电图信号的情绪识别的单次试验分类的原生规范模式匹配，”IEEE 生物医学工程汇刊，第 67 卷，第 8 期，第 2266-2275 页，2020 年 8 月。

[63] J. Jin, Z. Wang, R. Xu, C. Liu, X. Wang, and A. Cichocki, “Ro-
基于新型时间滤波器的 SSVEPs 检测的局部二值模式相似度测量，”IEEE 神经网络学习系统汇刊，待发表，doi: 10.1109/TNNLS.2021.3118468.

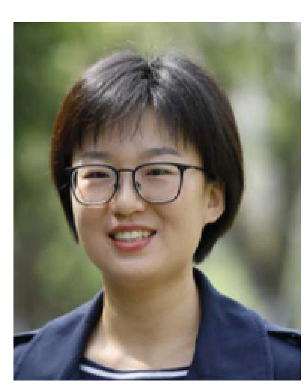
[64] V. D. M. 劳伦斯和 G. 希顿，“使用 t-SNE 可视化数据”，《机器学习杂志》 Learn. Res., 卷 9, 第 2605 期, 第 2579–2605 页, 2008.

周如霜目前在香港城市大学生物医学工程系攻读博士学位。他的研究兴趣包括情感计算和迁移学习。

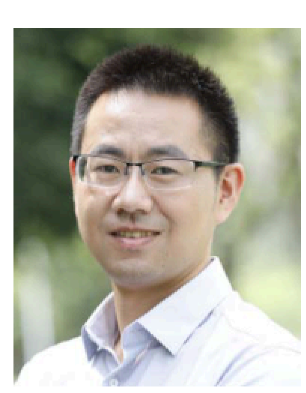
张志国（IEEE 会员）于 2000 年获得天津大学工学学士学位，2003 年获得中国科学技术大学理学硕士学位，2008 年获得香港大学哲学博士学位。他现任中国深圳哈尔滨工业大学计算机与智能学院教授。他的研究兴趣包括生物医学信号处理、神经工程和类脑计算。

冯红于 2007 年获得香港理工大学哲学博士学位。她现任香港教育大学数学与信息技术系助理教授。她的研究兴趣包括计算机视觉算法以及医疗和教育领域的人工智能。

张丽于 2017 年获得香港大学电气电子系哲学博士学位，并于 2018 年加入深圳大学医学院生物医学工程学院担任副研究员。她目前的研究兴趣主要集中在生物医学信号处理、数值优化和影像遗传学。

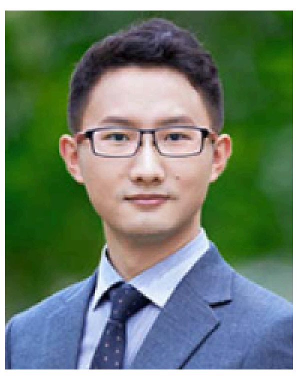


林玲玲于 2017 年获得中国科学院大学博士学位，随后在中国深圳大学进行博士后研究。她目前是深圳大学医学院生物医学工程学院的助理研究员。她目前的研究兴趣集中在脑电图神经反馈技术的发展和相关神经影像机制。



黄干于 2013 年获得上海交通大学博士学位，随后在比利时鲁汶天主教大学进行博士后研究。他目前是深圳大学医学院生物医学工程学院的助理教授。他在 Neuroimage、Neural Network 和 Neurocomputing 等期刊上发表了多篇高质量论文。他目前的研究兴趣集中在神经调节、脑机接口和神经假体。

Fali Li received the PhD degree from the University of Electronic Science and Technology of China, in 2020. He is now an associate researcher with the School of Life Science and Technology, Center for Information in Medicine, University of Electronic Science and Technology of China, China. His research interests include brain-computer interface, bio-informatics, and neural engineering.



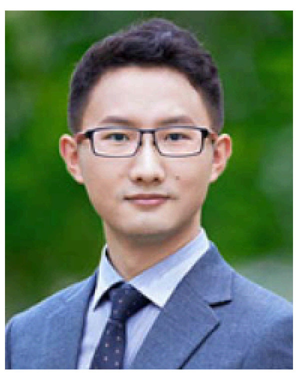
Xin Yang received from The Chinese University of Hong Kong, in 2019. He is now an assistant professor with the School of Biomedical Engineering, Shenzhen University, China. His research interests include medical image computing, intelligent ultrasound, cardiac image analysis and computer graphics.

Yining Dong received the BEng degree in electronic engineering from Tsinghua University, Beijing, China, in 2011, and the PhD degree in electrical engineering from the University of Southern California, Los Angeles, California, in 2016, She is currently an assistant professor with the School of Data Science, City University of Hong Kong, Hong Kong. Her research interests include process data analytics, statistical machine learning, smart manufacturing, and new material design.

Yuan-Ting Zhang (Fellow, IEEE) is currently the chairman and director of Hong Kong Center for Cerebro-cardiovascular Health Engineering and the chair professor with the City University of Hong Kong. He is a LRG member of Karolinska Institutet MWLC. He was the sensing system architect in Health Technology and Sensing Hardware Divisions at Apple Inc., California, the founding director of the Key Lab for Health Informatics of Chinese Academy of Sciences (CAS) and the founding director of CAS-SIAT Institute of Biomedical and Health Engineering. He dedicated his service to the Chinese University of Hong Kong from 1994 to 2015, where he served as the first head of the Division of Biomedical Engineering. He has been the editor-in-chief for IEEE Reviews in Biomedical Engineering since 2016 and chair of the IEEE 1708 Working Group. He was the editor-in-chief for *IEEE Transactions on Information Technology in Biomedicine* and the first editor-in-chief of *IEEE Journal of Biomedical and Health Informatics*. He served as vice preside of *IEEE Engineering in Medicine and Biology Society*. He was also the chair of 2016–2018 IEEE Award Committee in Biomedical Engineering and a member of IEEE Medal Panel for Healthcare Technology Award. His research interests include unobtrusive sensing and wearable devices, and neural muscular modeling. He was selected on the lists of China's Most Cited researchers by Elsevier and the top 2% researcher worldwide by Stanford University. He won a number of international awards including IEEEEMBS best journal paper awards, IEEE-EMBS Outstanding Service Award, IEEE-SA 2014 Emerging Technology Award, and Earl Owen Lecture with SMITIBEC2018 in Korea. He is elected as IAMBE fellow, AIMBE fellow and AAIA fellow for his contributions to the development of wearable and m-Health technologies.

Zhen Liang (Member, IEEE) received the PhD degree from The Hong Kong Polytechnic University, Hong Kong, in 2013. From 2012 to 2017, she was an algorithm development scientist with NeuroSky, Inc., Hong Kong. From 2018 to 2019, she was a specially-appointed assistant professor of the Graduate School of Informatics, Kyoto University, Japan. She is currently an Assistant Professor with the School of Biomedical Engineering, Medical School, Shenzhen University, China. Her current research interests include affective brain-computer interface, brain encoding and decoding systems, visual attention, and neural engineering.

Fali Li 于 2020 年获得电子科技大学博士学位。他目前是中国电子科技大学生命科学技术学院、医学信息中心的一名副研究员。他的研究兴趣包括脑机接口、生物信息学和神经工程。



Xin Yang 于 2019 年获得香港中文大学学位。他目前是中国深圳大学生物医学工程学院的一名助理教授。他的研究兴趣包括医学图像计算、智能超声、心脏图像分析和计算机图形学。

Yining Dong 于 2011 年在北京清华大学获得电子工程学士学位，并于 2016 年在美国南加州大学洛杉矶分校获得电气工程博士学位。她目前是香港城市大学数据科学学院的一名助理教授。她的研究兴趣包括过程数据分析、统计机器学习、智能制造和新材料设计。

张元庭（IEEE Fellow）目前担任香港脑心健康工程中心主任兼主任教授，同时也是香港城市大学的特任教授。他是瑞典卡罗琳斯卡学院 MWLC 的 LRG 成员。他曾担任苹果公司加利福尼亚州健康技术与传感硬件部门的传感系统架构师，是中国科学院健康信息重点实验室的创始主任，以及生物医学与健康工程研究所 CASSIAT 的创始主任。

他自 1994 年至 2015 年在中国大学（香港）服务，担任生物医学工程系首任系主任。自 2016 年起，他担任 IEEE 生物医学工程评论主编，并担任 IEEE 1708 工作组主席。他曾担任 IEEE 生物医学信息技术传输主编，以及 IEEE 生物医学与健康信息学期刊首任主编。他担任 IEEE 医学与生物学学会副主席。他还担任 2016-2018 年 IEEE 生物医学工程奖委员会主席，以及 IEEE 医疗技术奖奖评审团成员。他的研究兴趣包括非侵入式传感和可穿戴设备，以及神经肌肉建模。他被爱思唯尔评为中国高被引学者，并入选斯坦福大学全球前 2%顶尖学者名单。他获得多项国际奖项，包括 IEEEEMBS 最佳期刊论文奖、IEEE-EMBS 杰出服务奖、IEEE-SA 2014 新兴技术奖，以及在韩国 SMITIBEC2018 会议上获得 Earl Owen 讲座奖。 他因在可穿戴设备和 m-Health 技术发展方面的贡献，被选为 IAMBE Fellow、AIMBE Fellow 和 AAIA Fellow。

张峥（IEEE 会员）于 2013 年在香港理工大学获得博士学位。2012 年至 2017 年，她担任香港 NeuroSky 公司算法开发科学家。2018 年至 2019 年，她担任日本京都大学信息研究生院特聘副教授。目前，她是中国深圳大学医学院生物医学工程学院的助理教授。她当前的研究兴趣包括情感脑机接口、大脑编码与解码系统、视觉注意力和神经工程。